



Elastic–plastic response of a deep tunnel excavated in 3D Hoek–Brown rock mass considering different approaches for obtaining the out-of-plane stress

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ABSTRACT

The difficulty in the 3D analysis of a circular tunnel excavation lies in the selection of a reasonable softening equation for the out-of-plane stress (σ_z) in the plastic zone. This study developed and compared the semi-analytical solutions of a circular opening considering different approaches for obtaining σ_z , and the results were verified by the numerical results (1) The elastic approaches (e.g., generalized Hooke's law). cannot accurately reflect the mechanical properties of rock in the plastic zone and the results exhibit large differences compared with numerical results. (2) Plastic approaches demonstrate that the in-site axis stress (q) has a limited effect on the stability of surrounding rock, whereas the elastic approach-based results exhibit severe dependence on q . (3) The excavation responses based on the GZZ strength criterion and Hoek–Brown strength criteria have significant differences, reflecting the obvious effect of σ_z (σ_2) on the stability of rock mass, and an unreasonable criterion may result in inappropriate support measures (4) Different post-peak behaviors of rock mass (e.g., constitutive model, residual strength, dilatancy, etc.). significantly affect the surrounding rock stresses, and the applicability of elastic theories and shallow tunnel experiences need to be further considered in deep tunnel where the plastic zone is widely distributed.

1. Introduction

Analytical and numerical methods are two of the most effective approaches for studying the nonlinear mechanical response of deep tunnel excavation. Numerical methods have many advantages in the analysis of the 3D stress of the surrounding rock under irregular geometric boundary conditions, and can be used to study complex engineering problems, but they also have some limitations in parameter sensitivity analysis and evaluation of nonlinear mechanical properties. However, analytical or semi-analytical methods have some advantages in the stability design and control of the surrounding rock, parameter sensitivity analysis, and evaluating the internal correlation among parameters. Some scholars have introduced a mapping function to analyze tunnels excavated under asymmetric in-situ stress conditions as well as irregular geometric boundaries,^{1–4} but this is suitable for only elastic or viscoelastic analysis of shallow tunnels under low geo-stresses. However, for deep tunnels under high ground stress, the plastic deformation

of surrounding rock at post-peak stage is widely distributed,^{5–7} which is even larger than elastic deformation by more than a factor of ten.⁷ Elastic theory is therefore, practically restricted to shallow tunnels, and hence, elastic–plastic theory is essential and required for deep tunnel design and control. Although the assumptions of simple geometric boundary and hydrostatic stress state are often adopted in most elasto-plastic analytical or semi-analytical methods, it is useful to elucidate the non-linear behaviors of unloading surrounding rocks, conduct parameter research, and analyze the applicability of theoretical models,⁸ such as the interaction performance between viscoelastic–plastic surrounding rock and the support of deep tunnels considering the stress path,⁹ the relationship between rock mass dilatancy and deformation,^{10,11} rock mass rheological model and aging characteristics,¹² the applicability of the strength criterion,^{13,14} the difference between large and small deformation theories,^{15,16} and 3D space of the tunnel excavation and engineering design based on convergence–confinement method.^{17–19}

The excavation process causes stress redistribution and stress

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concentration around the tunnel. When the ratio of stress to strength exceeds a certain critical value, the surrounding rock exhibits plastic behavior, and the distribution of the plastic zone in the deep tunnel is more obvious, which is the most important reason for the large deformation of the tunnel. Meanwhile, the excavation causes the deep rock mass to present an obvious 3D stress state (i.e., $\sigma_1 > \sigma_2 > \sigma_3$),^{20–22} which brings significant difficulties to the analytical or semi-analytical solutions because of the unknown out-of-plane stress (axial stress, σ_z). For the plane strain model, the principal stress of the surrounding rock includes the in-plane stress (i.e., σ_r and σ_θ) and σ_z . For circular tunnels, many scholars have used 2D strength criteria (e.g., the Mohr–Coulomb criterion (MC) and HB strength criterion) to obtain the in-plane close-form or semi-analytical solutions based on different constitutive models, such as the elastic–perfectly plastic model (e-p-p),^{16,19,23} elastic–brittle–plastic model (e-b-p)^{24–27} and strain-softening model.^{11,28–34} However, these studies considered σ_θ and σ_r to be the maximum principal stress (σ_1) and minimum principal stress (σ_3), respectively. Their analytical process focused only on the in-plane mechanical properties, and the effects of σ_z (or σ_2) on the strength, deformation, and failure mode of the rock mass were neglected. However, σ_z is closely related to the in-plane stress, and σ_2 has a significant effect on stability and safety of the tunnel excavation. Therefore, this simple assumption would inevitably result in large errors, and the 2D strength criterion is often relatively conservative for the rock mass strength prediction.

The closed-form analytical solution based on the 2D strength criterion neglects the importance of the out-of-plane stress on rock strength and failure. Although the calculation process and analytical difficulty are greatly simplified, it does not reflect the 3D stress state of deep excavation. Therefore, it is considerably limited in terms of refined design and stability control of the surrounding rock. For a deep tunnel, excavation exhibits the obvious 3D stress states and nonlinear mechanical characteristics, and hence, reliable 3D and nonlinear theories should be used to study its mechanical behaviors, such as the 3D strength criterion and strain softening constitutive model. Some scholars used 3D strength criteria to study tunnel excavation; for instance, the viscoelastic–plastic solution of tunnel excavation based on the 3D HB criterion,³⁵ close-form analytical solution based on the Drucker–Prager (DP) criterion,³⁶ theoretical analysis of large deformation based on the generalized Zhang–Zhu (GZZ) criterion,¹⁵ and tunnel excavation response³⁷ and ground reaction or response curve (GRC)³² based on the unified strength criterion. It is often necessary to solve transcendental and nonlinear equations because of the stress invariants form of the 3D strength criterion, and its analytical difficulty is significantly more than that of the 2D criterion. Therefore, analysis results are mainly obtained using semi-analytical and numerical solutions. Meanwhile, σ_z is often assumed to be σ_2 , and hence, its applicability has been questioned. Some scholars have assumed that σ_z is the maximum, intermediate, and minimum stresses to obtain semi-analytical solutions, but the solution process increases the complexity of the problem to a certain extent.^{27,38,39}

In fact, both the in-plane and out-of-plane stresses can soften in the plastic zone compared with those in the elastic zone, and the softening curves are not independent of each other. Some scholars reported that the σ_z has an important effect on the in-plane stress and surrounding rock stability.^{39–41} Pan and Brown⁴⁰ noted that the σ_z has a significant effect on the surrounding rock stability and convergence deformation, and presented empirical relationships between the σ_z and in-plane stresses (σ_r and σ_θ) according to the numerical tests. Their empirical parameters are mainly related to the rock dilatancy, and the more obvious the dilatancy effect, the closer the σ_z is to the σ_θ , and a smaller σ_z often corresponds to a larger yield or failure zone of the surrounding rock. Wang et al.²⁷ found that the σ_z has a significant effect on the in-plane stress, deformation, and distribution of the plastic zone, and its influence degree is related to the difference between the in-plane and out-of-plane stresses, the difference between the peak and residual strength parameters, and the dilatancy characteristics of the rock mass.

Lu et al.³⁹ noted that there is a certain critical value of the in-situ out-of-plane stress (q_{cri}). The influence of q on the plastic radius is the same as the result obtained based on the plane strain model when $q < q_{cri}$, and the plastic radius increases with the increase in q when $q > q_{cri}$. These results show that the σ_z is closely related to the in-plane stress. However, although these studies quantitatively explained the effect of the σ_z on the stability of the surrounding rock using analytical or numerical approaches, they did not thoroughly investigate the approaches for obtaining σ_z , as well as its softening law in the plastic zone. In the 3D semi-analytical solution, the elastic Hooke's law is often applied to describe the softening law of σ_z , which is mainly suitable for the elastic plane strain model, and its applicability in the plastic zone is not considered. The elastic assumption and engineering experience from shallow tunnels may lead to large errors in the analysis of the plastic behaviors of rock mass at the post-peak stage, particularly for deep tunnels where the plastic zone is widely distributed.

This paper presents a semi-analytical solution for a deep circular tunnel excavated in a smooth GZZ rock mass considering different approaches for obtaining the σ_z . Different approaches were compared and verified through numerical results. And the effect of the strength criteria (HB and GZZ), constitutive models (e-b-p, e-p-p, and e-s-s), and dilatancy on the excavation response of deep surrounding rock are also quantitatively analyzed.

2. Rock mass models

2.1. GZZ strength criterion

The HB strength criterion is widely used for rock mass stability analysis. Its expression is⁴²

$$\sigma_1 = \sigma_3 + \sigma_c \left(m_b \frac{\sigma_3}{\sigma_c} + s \right)^a \quad (1)$$

where σ_c is the uniaxial compressive strength of the rock, and m_b , s , and a are the strength parameters, which can be obtained using Eq. (2).⁴²

$$m_b = m_i \exp \left(\frac{GSI - 100}{28 - 14D} \right) \quad (2a)$$

$$s = \exp \left(\frac{GSI - 100}{9 - 3D} \right) \quad (2b)$$

$$a = \frac{1}{2} + \frac{1}{6} [\exp(-GSI/15) - \exp(-20/3)] \quad (2c)$$

where m_i is the indoor rock parameter, GSI is the geological strength indicator, and D is the excavation disturbance coefficient.

Some scholars have developed different 3D HB criteria to overcome the limitation of original 2D HB criterion which has no consideration of σ_2 . Among these, the GZZ criterion (Eq. (3)) has the advantage of directly using the HB strength parameters and with high precision in fitting true triaxial experiment data and calculating practical tunnel excavation.^{43–46}

$$\frac{1}{\sigma_c^{(1/a-1)}} \left(\sqrt{3J_2} \right)^{1/a} + \frac{m_b}{2} \left(\sqrt{3J_2} \right) - m_b \sigma_{m,2} - s \sigma_c = 0 \quad (3)$$

where $\sigma_{m,2} = \frac{(\sigma_1 + \sigma_3)}{2}$, J_2 is the second invariant deviatoric stress, and $J_2 = S_{ij}S_{ji}/2$.

Cai et al.⁴⁷ modified the original GZZ criterion by replacing the term $\sigma_{m,2}$ in Eq. (3) with $\frac{1}{2} \left(I_1 - \frac{I_1 I_2 - 9I_3}{6J_2} \right)$ to obtain a new GZZ criterion with a smooth envelope on the π plane (referred to as smooth GZZ). Smooth GZZ is expressed in invariants as:

$$\frac{1}{\sigma_c^{(1/a-1)}} \left(\sqrt{3J_2} \right)^{1/a} + \frac{m_b}{2} \left(\sqrt{3J_2} \right) - \frac{m_b}{2} \left(I_1 - \frac{I_1 I_2 - 9I_3}{6J_2} \right) - s \sigma_c = 0 \quad (4)$$

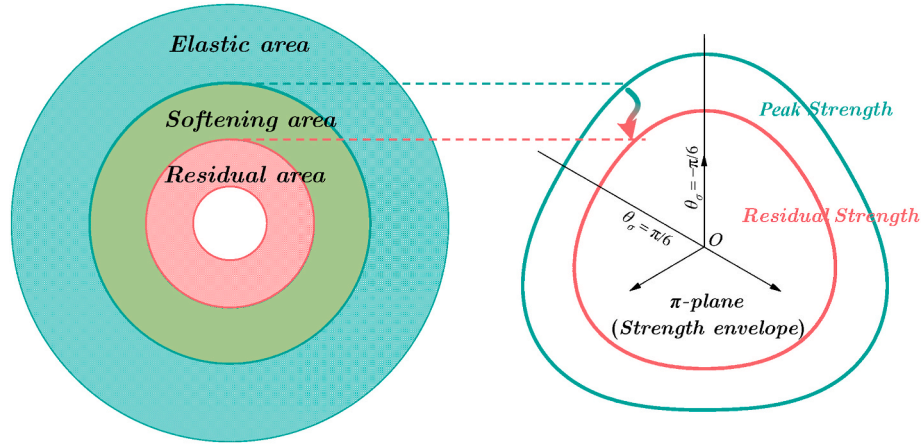


Fig. 1. Correspondence between surrounding rock divisions and rock mass parameters softening.

where θ_σ is the Lode angle and $\theta_\sigma = \tan^{-1} \left[\frac{2\sigma_2 - (\sigma_1 + \sigma_3)}{\sqrt{3}(\sigma_1 - \sigma_3)} \right]$, I_1 is the first invariant of the stress, $I_1 = \sigma_{ii}$.

The stress invariants and θ_σ satisfy the relationship:

$$\sin 3\theta_\sigma = -\frac{3\sqrt{3}}{2} \frac{J_3}{J_2^{3/2}} \quad (5a)$$

$$I_3 = -\frac{2}{3\sqrt{3}} J_2^{3/2} \sin 3\theta_\sigma - \frac{1}{3} I_1 J_2 + \frac{1}{27} I_1^3 \quad (5b)$$

So, the smooth GZZ can also be expressed in terms of stress invariants by introducing Eq. (5) into Eq. (4):⁴⁷

$$\frac{1}{\sigma_c^{(1/a-1)}} \left(\sqrt{3} J_2 \right)^{1/a} + \left(\frac{3 + \sin 3\theta_\sigma}{2\sqrt{3}} \right) m_b \sqrt{J_2} - m_b \frac{I_1}{3} - s\sigma_c = 0 \quad (6)$$

The normalized stress of σ_i^* can be represented as $\sigma_i^* = \frac{\sigma_i}{\sigma_c} + \frac{s}{m_b}$ to simplify the derivation. Eq. (6) can be simplified as:

$$\frac{1}{m_b} \left(\sqrt{3} J_2^* \right)^{1/a} + \left(\frac{3 + \sin 3\theta_\sigma^*}{2\sqrt{3}} \right) \sqrt{J_2^*} - \frac{I_1^*}{3} = 0 \quad (7)$$

where $\theta_\sigma^* = \tan^{-1} \left[\frac{2\sigma_2^* - (\sigma_1^* + \sigma_3^*)}{\sqrt{3}(\sigma_1^* - \sigma_3^*)} \right] = \theta_\sigma$, $I_1^* = \sigma_1^* + \sigma_2^* + \sigma_3^* = \frac{I_1}{\sigma_c} + \frac{3s}{m_b}$, $J_2^* = \frac{(\sigma_1^* - \sigma_2^*)^2 + (\sigma_2^* - \sigma_3^*)^2 + (\sigma_3^* - \sigma_1^*)^2}{6} = \frac{J_2}{\sigma_c^2}$, and $J_3^* = S_1^* S_2^* S_3^* = \left(\sigma_1^* - \frac{I_1^*}{3} \right) \left(\sigma_2^* - \frac{I_1^*}{3} \right) \left(\sigma_3^* - \frac{I_1^*}{3} \right) = \frac{J_3}{\sigma_c^3}$.

2.2. Strain softening constitutive model

The strain softening (s-s) constitutive model is widely used to represent the mechanical behavior of rock mass at the post-peak stage because of its unified mathematical form to reflect the e-b-p behavior of hard rock and e-p-p behavior of soft rock.^{10,48,49} This study mainly considers the weakening of the GSI . The initial GSI can be the peak GSI , that is, GSI_p . The plastic shear strain ($\gamma^p = \epsilon_1^p - \epsilon_3^p$) is the most important plastic internal variable that can be used as an independent variable for the $GSI(\gamma^p)$. In this study, it is assumed that the GSI follows a linear softening relationship with γ^p , as given in Eq. (8):²⁹

$$GSI = \begin{cases} GSI_p - \frac{GSI_p - GSI_r}{\gamma^{p,*}} \gamma^p & (0 < \gamma^p < \gamma^{p,*}) \\ GSI_r & (\gamma^p \geq \gamma^{p,*}) \end{cases} \quad (8)$$

where $\gamma^{p,*}$ is the maximum plastic shear strain.

Eq. (8) has also been adopted by many scholars. And GSI_r is the residual GSI , which can be calculated using Eq. (9).⁴⁸

$$GSI_r = GSI_p \exp(-0.0134 GSI_p) \quad (9)$$

Excavation in deep rock mass shows significant dilatancy and post-peak behavior. Cai et al. therefore developed a dilatancy model to comprehensively reflects the influences of confining pressure (σ_3) and γ^p on nonlinear post-peak behavior, with the advantages of a simple mathematical form, while requiring fewer parameters and demonstrating a clear physical significance. It can be expressed as⁴⁵

$$\frac{\psi(\sigma_3, \gamma^p)}{\varphi} = \frac{1}{\lg(k)} \lg \left(\frac{\sigma_3}{\sigma_{ci}} + k \right) e^{-\frac{\gamma^p}{\varphi \sigma_{ci}}} \quad (10)$$

where k is a rock-related coefficient, and can be approximately taken as 0.0041 for sedimentary (soft) rocks and 0.0177 for crystalline (hard) rocks.

The rock mass parameters (i.e., m_b , s , and a) can be calculated from Eq. (2), and the dilatancy angle can be obtained using Eq. (10), which is indirectly related to γ^p . The surrounding rock could be divided into three parts according to the GSI values, i.e., elastic area, softening area, and residual area. This simplification facilitates the calculation process and make acceptable error (see Fig. 1).

3. Mechanical response of circular tunnel excavation

Excavation in deep rock mass produces wide distribution of plastic zone. The exact solution of mechanical response for non-circular tunnel with asymmetric loads is very difficult, and even can only be solved numerically. Therefore, this paper uses the finite difference method to solve the excavation elastic-plastic mechanical response of circular tunnels under symmetric loads.

3.1. Excavation responses of surrounding rock in the elastic zone

The rock stress in the elastic zone of a circular opening is⁵⁰

$$\sigma_\theta = p + (p - p_{cr})(r_{cr}/r)^2 \quad (11a)$$

$$\sigma_r = p - (p - p_{cr})(r_{cr}/r)^2 \quad (11b)$$

$$\sigma_z = \nu(\sigma_\theta + \sigma_r) + (q - 2\nu p) = q \quad (11c)$$

where r is the radius of the surrounding rock to the tunnel center, r_{cr} is the radius of the elastic-plastic boundary and $r_{cr} = r_0$ (i.e., tunnel radius) when the rock is in the elastic stress state, p is the far-field in-plane stress, p_{cr} is the radial stress at the elastic-plastic boundary and $p_{cr} = p_i$ (i.e., support pressure) when the rock is in the elastic stress state, p_i is the support stress, ν is the Poisson's ratio, and q is the in-situ out-of-plane stress.

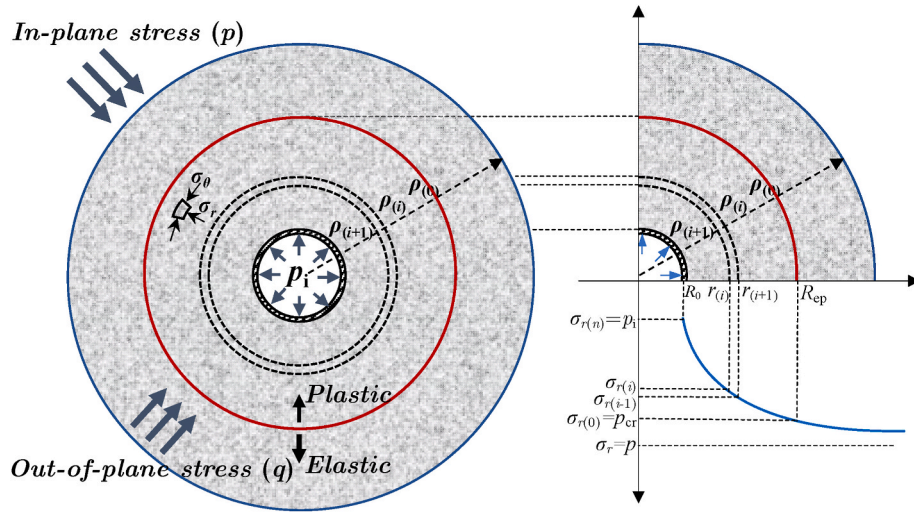


Fig. 2. Schematic diagram of stress equalizing circle.

The strain (ϵ_r , ϵ_z and ϵ_θ) and radial displacement (u_r) of the surrounding rock is

$$\epsilon_\theta = -\epsilon_r = (p - p_{cr})r_{cr}^2 / (2Gr^2) \quad (12a)$$

$$\epsilon_z = 0 \quad (12b)$$

$$u_r = (p - p_{cr})r_{cr}^2 / (2Gr) \quad (12c)$$

The key to solving the elastic stress is to determine the r_{cr} , which should be obtained according to the plastic stress state discussed later.

3.2. Excavation responses of surrounding rock in the plastic zone

Rock in the plastic region may exhibit strain softening, perfectly plastic, or brittle-plastic behaviors. It is difficult to obtain a unified closed-form analytical solution because of the nonlinear and complex stress-strain constitutive of the post-peak rock mass behaviors, and a semi-analytical solution seems to be an effective approach. Therefore, this study used the finite difference method to obtain a semi-analytical solution. The rock mass in the plastic zone can be divided into n rings, and the stress, strain, and displacement of the i -th ($0 < i < n$) circular ring can be computed by treating the response of the $(i-1)$ -th circular ring as its boundary condition. This idea can also refer to the original research conducted by Lee and Pietruszczak (2008).²⁹

The plastic zone was divided equally into n rings according to the radial stress, and the ring radius was normalized (Fig. 2) as

$$\rho_{(i)} = r_{(i)} / r_{cr} \quad (13)$$

The elastic-plastic boundary is defined as the outer boundary of the first ($i = 1$) ring, that is, $\rho_{(0)} = 1$; $\rho_{(n)} = \frac{r_0}{r_{cr}}$ is for the n -th circular ring at the tunnel wall. The stress increment between the two adjacent rings is set to $\Delta\sigma_r = \sigma_{r(i)} - \sigma_{r(i-1)} = -\frac{p_{cr}-p_i}{n}$.

3.2.1. Stresses of the elastic-plastic boundary

From Eq. (11), we can obtain the stresses in the elastic-plastic boundary for $r = r_{cr}$ or $\rho_{(0)} = 1$.

$$\sigma_{r(0)} = p_{cr} \quad (14a)$$

$$\sigma_{\theta(0)} = 2p - p_{cr} \quad (14b)$$

$$\sigma_{z(0)} = q \quad (14c)$$

$$\epsilon_{r(0)} = (p_{cr} - p) / (2G) \quad (14d)$$

$$\epsilon_{\theta(0)} = (p - p_{cr}) / (2G) \quad (14e)$$

Eq. (14) satisfies the elastic and plastic yield equations (smooth GZZ strength criterion) according to the stress continuity condition. Substituting Eq. (14) into Eq. (7), we obtain

$$\frac{1}{m_{b(0)}} \left(\sqrt{3J_{2(0)}^*} \right)^{1/a_{(0)}} + \left(\frac{\sqrt{3J_{2(0)}^*}}{2} - \frac{3J_{3(0)}^*}{4J_{2(0)}^*} \right) - \frac{I_{1(0)}^*}{3} = 0 \quad (15)$$

It can be determined that $I_{1(0)}^* = \frac{I_{1(0)}}{\sigma_c} + \frac{3s}{m_b}$, $J_{2(0)}^* = \frac{J_{2(0)}}{\sigma_c^2}$, and $J_{3(0)}^* = \frac{J_{3(0)}}{\sigma_c^3}$, where $I_{1(0)}^* = I_{1(0)}^*(p_{cr})$, $J_{2(0)}^* = J_{2(0)}^*(p_{cr})$, and $J_{3(0)}^* = J_{3(0)}^*(p_{cr})$ are all functions of p_{cr} , which can be solved via iteration. The principal stress at the elastic-plastic boundary can then be obtained.

It should be noted that many studies considered the out-of-plane stress σ_z to be σ_2 in the plastic zone; however, this assumption is not necessary if a smooth GZZ is used in the semi-analytical study because of the symmetrical location of the principal stresses in Eq. (6). In other words, σ_z is not necessarily required to be σ_2 and can be an arbitrary principal stress in the calculation. Regardless of the distribution of the out-of-plane stress and in-plane stress, the value of $\sin 3\theta_\sigma = -\frac{3\sqrt{3}}{2} \frac{J_3}{J_2^{3/2}}$ always fixed. So, there is no tedious and repetitive derivation process for a case-by-case analysis based on the relative values of σ_z , σ_r , and σ_θ if a smooth GZZ is used, and its smoothness capability is highly suitable for numerical or semi-analytical calculation.

3.2.2. Stress continuity at the elastic-plastic boundary

In the elastic-plastic zone, the strength criterion satisfies

$$F(\sigma_{(i)}, S_{a(i)}) = \begin{cases} < 0, & r < r_{cr} (\rho_{(0)} > 1) \\ 0^-, & r = r_{cr} (\rho_{(0)} = 1^+) \\ 0, & r \geq r_{cr} (\rho_{(0)} \leq 1) \end{cases} \quad (16)$$

where $\sigma_{(i)} = [\sigma_{r(i)}, \sigma_{\theta(i)}, \sigma_{z(i)}]$ is the stress vector of the i -th rock ring, and S_a is the strength parameter, i.e., m_b , s , and a .

According to Eq. (16), the strength criterion is a continuous function in the elastic-plastic zone. The stress in the elastic zone is inside the strength envelope with $F < 0$; in the elastic zone and near the elastic-plastic boundary, $F = 0^-$, whereas in the elastic-plastic boundary and plastic zone, $F = 0$.

It is usually assumed that σ_θ and σ_r are σ_1 and σ_3 when the HB criterion is used in previous analytical methods, and Eq. (1) can be rewritten as

$$\sigma_{\theta(i)}^* = \sigma_{r(i)}^* + \left(m_{b(i)} \sigma_{r(i)}^* \right)^{a(i)} \quad (17)$$

Substituting $\sigma_{r(0)}^* = p_{cr}^*$ and $\sigma_{\theta(0)}^* = 2p^* - p_{cr}^*$ into Eq. (17), we obtain

$$\left(m_{b(0)} p_{cr}^* \right)^{a(0)} + 2p_{cr}^* - 2p^* = 0 \quad (18)$$

Solving Eq. (18) yields the radial stress ($\sigma_{r(0)}^* = p_{cr}^*$) at the elastic–plastic boundary, and $\sigma_{\theta(i)}^*$ can be obtained by substituting $\sigma_{r(i)}^* = \sigma_{r(i-1)}^* + \Delta\sigma_r^*$ into Eq. (17).

However, the HB-based analytical solution neglects σ_z , and σ_z is not always equal to σ_2 in practice. Therefore, the assumption that σ_r and σ_θ are σ_3 and σ_1 would result in a limited application.

However, for the smooth GZZ-based stresses calculation, there are two unknown stresses (i.e., σ_θ and σ_z) to be computed (σ_r is known, and $\sigma_{r(i)}^* = \sigma_{r(i-1)}^* + \Delta\sigma_r^*$). Therefore, another independent equation needs to be formulated in addition to the strength criterion, and the added equation should accurately reflect the relationship between the in-plane stress and the out-of-plane stress, which is given as

$$G(\sigma_{(i)}, \mathbf{c}_{(i)}) = 0 \quad (19)$$

where $\mathbf{c}_{(i)} = [c_{r(i)}, c_{\theta(i)}, c_{z(i)}]$ is related parameters.

Because the strength criterion is a continuous function, [supplementary Eq. \(19\)](#) is therefore required to satisfy the stress continuity condition. The sufficient and necessary conditions of Eq. (19) can be written as

$$G_{(i)} = G(\sigma_{(i)}, \mathbf{c}_{(i)}) = 0 \Leftrightarrow \begin{cases} G_{(0)} = G(\sigma_{(0)}, \mathbf{c}_{(0)}) = 0 \\ dG_{(i)} = \Delta G_{(i)} = 0 \end{cases} \quad (20)$$

Here, $\sigma_{(0)} = [\sigma_{r(0)}, \sigma_{\theta(0)}, \sigma_{z(0)}] = (p_{cr}, 2p - p_{cr}, q)$.

Similarly, the strength criterion in terms of continuity equation can be written as

$$F_{(i)} = F(\sigma_{(i)}, S_{a(i)}) = 0 \Leftrightarrow \begin{cases} F_{(0)} = F(\sigma_{(0)}, GSI_0) = 0 \\ \Delta F_{(i)} = \frac{\partial F_{(i)}}{\partial \sigma} \Delta \sigma_{(i)} + \frac{\partial F_{(i)}}{\partial S_a} \Delta S_{a(i)} = 0 \end{cases} \quad (21)$$

The strength parameter $S_a = GSI(\gamma^p)$ and the term $\frac{\partial F_{(i)}}{\partial S_a} \Delta S_{a(i)}$ in Eq. (21) can be written as

$$\frac{\partial F_{(i)}}{\partial S_a} \Delta S_a = \sum_{k=1}^3 \left(\frac{\partial F_{(i)}}{\partial P_k} \right) \Delta P_k, \quad (P_1, P_2, P_3 = m_b, s, a) \quad (22)$$

Here, $\frac{\partial F_{(i)}}{\partial P_k}$ is the partial derivative of the strength criterion with respect to the strength parameter (m_b , s , and a).

For a smooth GZZ, $\frac{\partial F}{\partial \sigma_{ij}}$ can be written as⁴⁷

$$\frac{\partial F}{\partial \sigma_{ij}} = \frac{\partial Q}{\partial I_1} \frac{\partial I_1}{\partial \sigma_{ij}} + \frac{\partial Q}{\partial \sqrt{J_2}} \frac{\partial \sqrt{J_2}}{\partial \sigma_{ij}} + \frac{\partial Q}{\partial J_3} \frac{\partial J_3}{\partial \sigma_{ij}} \quad (23)$$

Here,

$$\frac{\partial Q}{\partial I_1} = -\frac{m_b}{3} \quad (24a)$$

$$\frac{\partial Q}{\partial \sqrt{J_2}} = g(J_2, \theta_\sigma) m_b \quad (24b)$$

$$\frac{\partial Q}{\partial J_3} = -\frac{3m_b}{4J_2} \quad (24c)$$

$$\frac{\partial I_1}{\partial \sigma_{ij}} = \delta_{ij} \quad (24d)$$

$$\frac{\partial \sqrt{J_2}}{\partial \sigma_{ij}} = \frac{S_{ij}}{2\sqrt{J_2}} \quad (24e)$$

$$\frac{\partial J_3}{\partial \sigma_{ij}} = S_{ik} S_{kj} - \frac{2J_2}{3} \delta_{ij} \quad (24f)$$

where $g(J_2, \theta_\sigma) = \sqrt{3} \left[\frac{(\sqrt{3J_2})^{\frac{1}{a}-1}}{a m_b \sigma_c} + \frac{\sqrt{3J_2}}{2J_2^{3/2}} + \frac{1}{2} \right]$.

Substituting Eq. (24) into (23), we obtain

$$\frac{\partial F}{\partial \sigma_{ij}} = \frac{m_b}{2} \left(\frac{1}{3} \delta_{ij} + \frac{g(J_2, \theta_\sigma) S_{ij}}{\sqrt{J_2}} - \frac{3S_{ik} S_{kj}}{2J_2} \right) \quad (25)$$

According to Eq. (6), $\frac{\partial F_{(i)}}{\partial S_a}$ can be evaluated as:

$$\frac{\partial F}{\partial P_k} = \begin{cases} \frac{\partial F}{\partial m_b} = \frac{\sqrt{3J_2}}{2} - \frac{3J_3}{4J_2} - \frac{I_1}{3} \\ \frac{\partial F}{\partial s} = -\sigma_c \\ \frac{\partial F}{\partial a} = \frac{(\sqrt{3J_2})^{1/a}}{a^2 \sigma_c^{(1/a-1)}} \ln \left(\frac{\sigma_c}{\sqrt{3J_2}} \right) \end{cases} \quad (26)$$

For soft rock or rock in the plastic flow stage, it is appropriate to use a perfectly plastic model with constant strength parameters. $dS_a = 0$, and hence, a linear equation $dF_{(i)} = \left(\frac{\partial F_{(i)}}{\partial \sigma} \right) d\sigma = 0$ is obtained based on Eq. (21), which significantly improves the calculation efficiency.

The stress components in the plastic zone can be obtained by solving Eqs. (20) and (21).

3.2.3. Supplementary equation of the out-of-plane stress in the plastic zone

Although the finite difference method is widely used in semi-analytical solution of circular tunnel excavation, the out-of-plane stress is often overlooked or wrongly assumed. This part will present four supplementary equations $G^k = 0$ ($k = \text{I, II, III, IV}$), as reported in the literature and in this paper. The characteristics of various approaches and their effects on the stress components in the plastic zone were compared and analyzed, and a unified softening equation of the out-of-plane stress was formulated.

3.2.3.1. Approach I (elastic approach). Eq. (27) can be obtained for the elastic plane strain problem based on the Hooke's law, $\varepsilon_z = \frac{[\sigma_z - \nu(\sigma_\theta + \sigma_r)]}{E} = 0$.

$$G_{(i)}^I = \sigma_{z(i)} - \nu(\sigma_{\theta(i)} + \sigma_{r(i)}) = 0 \quad (27)$$

Xu and Xia (2021)¹⁵ used the elastic approach (Eq. (27)) to obtain a semi-analytical solution by expressing GZZ in the form of σ_θ and σ_r . However, $\sigma_z = \nu(\sigma_\theta + \sigma_r)$ is indicated as the lateral stress owing to the Poisson effect under the condition of no initial stress loading, it is not suitable for the arbitrary in-situ out-of-plane stress (q).

At the elastic–plastic boundary, $\sigma_{z(0)} = 2\nu p$, which cannot satisfy the stress continuity condition because $2\nu p \neq q$. Therefore, this elastic approach is only limited to the special case of $q = \nu(\sigma_\theta + \sigma_r)$.

3.2.3.2. Approach II (elastic approach). Some researchers^{16,27,40,51} divided the axis strain (ε_z) into elastic strain (ε_z^e) and plastic strain (ε_z^p), and obtained $\sigma_{z(i)}$ in the plastic zone using the generalized Hooke's law under the assumption that $\varepsilon_z = 0$, as given in Eqs. (28) and (29), respectively.

$$\varepsilon_z = \varepsilon_z^e + \varepsilon_z^p = 0 \quad (28a)$$

$$\varepsilon_z^e = \frac{1}{E} [\sigma_z - \nu(\sigma_r + \sigma_\theta) - (1 - 2\nu)q] \quad (28b)$$

$$\sigma_{z(i)} = \nu(\sigma_{\theta(i)} + \sigma_{r(i)}) + (q - 2\nu p) - E\varepsilon_{z(i)}^p \quad (29)$$

This approach provides a good reference for solving $\sigma_{z(i)}$ in the plastic

zone. Finite element results obtained in several studies have demonstrated that ε_z^p is almost zero in the plane strain model, which is much smaller than ε_r^p and ε_θ^p . Therefore, Eq. (29) can also be written as

$$G_{(i)}^{\text{II}} = \sigma_{z(i)} - [\nu(\sigma_{\theta(i)} + \sigma_{r(i)}) + (q - 2\nu p)] = 0 \quad (30)$$

By comparing Eqs. (27) and (30), it can be observed that there is only a constant difference $(q - 2\nu p)$ between $G_{(i)}^{\text{II}} = 0$ and $G_{(i)}^{\text{I}} = 0$. Moreover, $G_{(i)}^{\text{II}} = 0$ can satisfy the stress continuity condition because $\sigma_{z(0)} = q$ is obtained at the elastic-plastic boundary.

$G_{(i)}^{\text{II}} = 0$ is based on the generalized Hooke's law, which is derived from the foundation of linear elasticity theory. Eq. (30) can also be expressed in the form of an incremental stress, which is given as¹⁴

$$\begin{bmatrix} \Delta \varepsilon_{r(i)}^e \\ \Delta \varepsilon_{\theta(i)}^e \\ \Delta \varepsilon_{z(i)}^e \end{bmatrix} = \frac{1}{E} \begin{bmatrix} 1 & -\nu & -\nu \\ -\nu & 1 & -\nu \\ -\nu & -\nu & 1 \end{bmatrix} \begin{bmatrix} \Delta \sigma_r \\ \Delta \sigma_{\theta(i)} \\ \Delta \sigma_{z(i)} \end{bmatrix} \quad (31)$$

The incremental expression of the out-of-plane stress ($\Delta \sigma_{z(i)}$) for the plane strain problem can then be written as

$$\Delta \sigma_{z(i)} - \nu(\Delta \sigma_r + \Delta \sigma_{\theta(i)}) = 0 \quad (32)$$

Eqs. (30) and (32) would be equivalent if $\sigma_{z(0)} = q$ is satisfied, and Eq. (32) can also be obtained from $\Delta G_{(i)}^{\text{II}} = G_{(i)}^{\text{II}} - G_{(i-1)}^{\text{II}} = 0$. This is from linear superposition of the elastic Hooke's law.

3.2.3.3. Approach III (dilatancy-dependence approach). Pan and Brown⁴⁰ combined finite element numerical results to obtain the statistical equation of σ_z in the plastic zone for the plane strain model

$$\sigma_{z(i)} = \begin{cases} \frac{\sqrt{3}(k' + 2)}{k' - 1} \alpha_{\text{MC}} \left(\frac{\sigma_{\theta(i)} - \sigma_{r(i)}}{2} \right) + \left(\frac{\sigma_{\theta(i)} + \sigma_{r(i)}}{2} \right), \text{MC} \\ \frac{n_{\text{HB}}}{m'} \left(\frac{\sigma_{\theta(i)} - \sigma_{r(i)}}{2} \right) + \left(\frac{\sigma_{\theta(i)} + \sigma_{r(i)}}{2} \right), \text{HB} \end{cases} \quad (33)$$

where k' and m' are the residual strength parameters based on MC and HB, respectively; α_{MC} and n_{HB} are the dilatancy correction parameters based on the plastic potential functions of MC and HB, respectively.

Deng et al.⁵² obtained the analytical equation of the plane strain model in the plastic zone through uniform matching of the relationship between MC and Mises criteria:

$$\sigma_{z(i)} = \frac{1 - \sin \psi_{(i)}}{2} \sigma_{\theta(i)} + \frac{1 + \sin \psi_{(i)}}{2} \sigma_{r(i)} \quad (34)$$

Liu et al.⁵³ formulated an equation similar to Eq. (34) by matching the relationship between MC and DP criteria and obtained a mathematical relationship given as follows:

$$\sigma_{z(i)} = \frac{1 + \sin \psi_{(i)}}{2} \sigma_{\theta(i)} + \frac{1 - \sin \psi_{(i)}}{2} \sigma_{r(i)} \quad (35)$$

Eqs. (33)–(35) can be written uniformly as

$$G_{(i)}^{\text{III}} = \sigma_{z(i)} - \left\{ \frac{\sigma_{\theta(i)}}{2} [1 + f(\psi)] + \frac{\sigma_{r(i)}}{2} [1 - f(\psi)] \right\} = 0 \quad (36)$$

$G_{(i)}^{\text{III}} = 0$ indicates that the out-of-plane stress in the plastic zone is affected by rock mass dilatancy. Poor quality rock mass ($\text{GSI} < 25$) follows the constant volume flow rule,⁴⁹ and it is considered that the surrounding rock excavation process does not produce volume dilation with $\psi = 0$ and $\sigma_z = \frac{(\sigma_\theta + \sigma_r)}{2}$. Meanwhile, the dilatancy angle of engineering rock mass is generally 0–0.25 times the internal friction angle (φ),⁴⁹ and $\frac{f(\psi)(\sigma_{\theta(i)} - \sigma_{r(i)})}{2}$ in Eq. (36) is much smaller than $\frac{(\sigma_\theta + \sigma_r)}{2}$. $G_{(i)}^{\text{III}} = \sigma_{z(i)} - \frac{1}{2}(\sigma_{\theta(i)} + \sigma_{r(i)}) = 0$ can, therefore, be used as an approximate equation of Eq. (36).

3.2.3.4. Approach IV (plasticity approach). In classical elastoplastic

mechanics, the plastic strain increment in each principal stress direction is related to the adopted plastic potential function Q , that is,

$$d\varepsilon_{ij}^p = d\lambda \frac{\partial Q}{\partial \sigma_{ij}} = d\lambda \left(\frac{\partial Q}{\partial I_1} \frac{\partial I_1}{\partial \sigma_{ij}} + \frac{\partial Q}{\partial \sqrt{J_2}} \frac{\partial \sqrt{J_2}}{\partial \sigma_{ij}} + \frac{\partial Q}{\partial J_3} \frac{\partial J_3}{\partial \sigma_{ij}} \right) \quad (37)$$

Smooth GZZ associated flow rule is considered; that is, $F = Q$ was applied in this study. The plastic flow vector is obtained using Eq. (25)

$$d\varepsilon_{ij}^p = d\lambda \frac{m_b}{2} \left(\frac{1}{3} \delta_{ij} + \frac{g(J_2, \theta_\sigma) S_{ij}}{\sqrt{J_2}} - \frac{3S_{ik}S_{kj}}{2J_2} \right) \quad (38)$$

For a plane strain model, the axial strain is zero (i.e., $d\varepsilon_z = 0$), that is,

$$d\varepsilon_z = d\varepsilon_z^e + d\varepsilon_z^p = 0 \quad (39)$$

The deformation caused by deep tunnel excavation under high ground stress is mainly plastic deformation, and the elastic strain is very small compared with the plastic strain. Therefore, Eq. (39) can also be approximated as⁵⁴

$$d\varepsilon_z = d\varepsilon_z^e + d\varepsilon_z^p \approx d\varepsilon_z^p = 0 \quad (40)$$

Combining Eqs. (39) and (38), we obtain

$$d\varepsilon_z^p = d\lambda \frac{m_b}{2} \left(\frac{g(J_2, \theta_\sigma) S_z}{\sqrt{J_2}} - \frac{3S_z^2}{2J_2} + \frac{1}{3} \right) = 0 \quad (41)$$

Therefore, $G_{(i)}^{\text{V}} = 0$ can be written as

$$G_{(i)}^{\text{IV}} = \frac{S_{z(i)}}{\sqrt{J_{2(i)}}} g(J_{2(i)}, \theta_{\sigma(i)}) - \frac{3S_{z(i)}^2}{2J_{2(i)}} + \frac{1}{3} = 0 \quad (42)$$

It should be noted that $S_{z(i)} = \sigma_{z(i)} - \frac{I_{1(i)}}{3} = 0$ is satisfied when $\sigma_z = \frac{(\sigma_\theta + \sigma_r)}{2}$, and Eq. (42) can be almost satisfied if a small stress increment is added to $\sigma_z = \frac{(\sigma_\theta + \sigma_r)}{2}$. Thus, $\sigma_z = \frac{(\sigma_\theta + \sigma_r)}{2}$ can also serve as the approximate solution of $G_{(i)}^{\text{IV}} = 0$. It can be inferred that the semi-analytic results based on $G_{(i)}^{\text{V}} = 0$ and $G_{(i)}^{\text{III}} = 0$ are similar.

3.2.4. Unified softening equation of the out-of-plane stress

As presented above, the relationship between the out-of-plane stress and the in-plane stress can be expressed as

$$\sigma_z = f(\sigma_\theta + \sigma_r) + g(\sigma_\theta - \sigma_r) \quad (43)$$

A linear equation can be assumed for simplicity, and Eq. (43) can be written as

$$\sigma_z = c_1(\sigma_\theta + \sigma_r) + c_2(\sigma_\theta - \sigma_r) + c_0 = c_r \sigma_r + c_\theta \sigma_\theta + c_0 \quad (44)$$

where $c_r = c_1 - c_2$ and $c_\theta = c_1 + c_2$.

According to Eq. (20), the unknown parameter c_0 is obtained by introducing the stress components at the elastic-plastic boundary, that is, $\sigma_{(0)} = [\sigma_{r(0)}, \sigma_{\theta(0)}, \sigma_{z(0)}] = (p_{\text{cr}}, 2p - p_{\text{cr}}, q)$, into Eq. (44). Then, substituting c_0 back into Eq. (44), we obtain

$$\sigma_z = q + c_r(\sigma_r - p_{\text{cr}}) + c_\theta[\sigma_\theta - (2p - p_{\text{cr}})] \quad (45)$$

In Eq. (44), $\sigma_r - p_{\text{cr}}$ and $\sigma_\theta - (2p - p_{\text{cr}})$ can be regarded as the softening stress of σ_r and σ_θ in the plastic zone, respectively, and c_r and c_θ can be regarded as the corresponding stress softening coefficients. Therefore, $G_{(i)}^{\text{V}} = 0$ can be written as

$$G_{(i)}^{\text{V}} = \sigma_{z(i)} - \{q + c_r(\sigma_{r(i)} - p_{\text{cr}}) + c_\theta[\sigma_{\theta(i)} - (2p - p_{\text{cr}})]\} = 0 \quad (46)$$

Substituting $G_{(i)}^{\text{II}} = 0$ into Eq. (30), we obtain

$$G_{(i)}^{\text{II}} = \sigma_{z(i)} - \{q + \nu(\sigma_{r(i)} - p_{\text{cr}}) + \nu[\sigma_{\theta(i)} - (2p - p_{\text{cr}})]\} = 0 \quad (47)$$

A comparison of $G_{(i)}^{\text{V}} = 0$ and $G_{(i)}^{\text{II}} = 0$ (Eq. (46) and (47)) shows that $G_{(i)}^{\text{II}} = 0$ is a special case of $G_{(i)}^{\text{V}} = 0$ when $c_r = c_\theta = \nu$. The semi-analytic results obtained using Eq. (45) agree well with the numerical results.

Table 1
 $c_{r(i)}$ and $c_{\theta(i)}$ for different approaches obtaining out-of-plane stress.

Approaches	Mathematical form	$\Delta\sigma_{z(i)} = (c_{r(i)}\Delta\sigma_{r(i)} + c_{\theta(i)}\Delta\sigma_{\theta(i)})$	
		$c_{r(i)}$	$c_{\theta(i)}$
$G_{(i)}^I = 0$	$\sigma_z = \nu(\sigma_\theta + \sigma_r)$	ν	ν
$G_{(i)}^{II} = 0$	$\sigma_z = \nu(\sigma_\theta + \sigma_r) + (q - 2\nu p)$	ν	ν
$G_{(i)}^{III} = 0$	$\sigma_z = \frac{\sigma_\theta}{2}[1 + f(\psi)] + \frac{\sigma_r}{2}[1 - f(\psi)]$	$\frac{1 - f(\psi)}{2}$	$\frac{1 + f(\psi)}{2}$
$G_{(i)}^{IV} = 0$	$\frac{S_z}{\sqrt{J_2}}g(J_2, \theta_\sigma) - \frac{3S_z^2}{2J_2} + \frac{1}{3} = 0$	$\approx \frac{1}{2}$	$\approx \frac{1}{2}$

$c_r = c_\theta = \nu$ can be used as an accurate expression of the out-of-plane stress in the elastic zone, whereas the difference between c_r and c_θ is large in the plastic zone, and it seems that the Poisson's ratio (ν) is separated into two coefficients. In fact, the Poisson effect is directly affected by rock mass dilatancy, and the dilatancy in the plastic zone is related to the stress state and plastic deformation process.^{10,55} Therefore, there will be a large difference in the out-of-plane stresses between semi-analytical results and numerical results if a constant value is used for ν in the plastic calculation.

According to the expression in Eq. (20), $G_{(i)} = 0$ is equivalent to the equations of $G_{(0)} = G(\sigma_{(0)}, c_{(0)}) = 0$ and $\Delta G_{(i)} = G_{(i)} - G_{(i-1)} = 0$ in the

finite difference method used in this study. Therefore, Eq. (46) can also be written as

$$\sigma_{z(0)} = q \quad (48a)$$

$$\Delta G(\sigma_{(i)}, c_{(i)}) = \Delta\sigma_{z(i)} - (c_{r(i)}\Delta\sigma_{r(i)} + c_{\theta(i)}\Delta\sigma_{\theta(i)}) = 0 \quad (48b)$$

Eq. (48) is the incremental expression of the softening equation of the out-of-plane stress $G(\sigma_{(i)}, c_{(i)}) = 0$ in the plastic zone, which does not depend on the concrete mathematical expression. Eq. (48) can be transformed into any of the out-of-plane stress expressions presented in Section 3.2.3 by selecting the appropriate $c_{r(i)}$ and $c_{\theta(i)}$, as presented in Table 1.

Furthermore, $\sigma_{(0)} = [\sigma_{r(0)}, \sigma_{\theta(0)}, \sigma_{z(0)}] = (p_{cr}, 2p - p_{cr}, q)$ is also the solution of Eq. (21), i.e., $F_{(0)} = F(\sigma_{(0)}, GSI_0) = 0$. Therefore, the determined equations for solving the equations of $F_{(i)} = 0$ and $\Delta G_{(i)} = 0$ in the plastic zone are

$$[\sigma_{r(0)}, \sigma_{\theta(0)}, \sigma_{z(0)}] = (p_{cr}, 2p - p_{cr}, q) \quad (49a)$$

$$\sum_{k=r,\theta,z}^3 \frac{\partial F_{(i)}}{\partial \sigma_k} \Delta\sigma_k + \frac{\partial F_{(i)}}{\partial S_a} \Delta S_a = 0 \quad (49b)$$

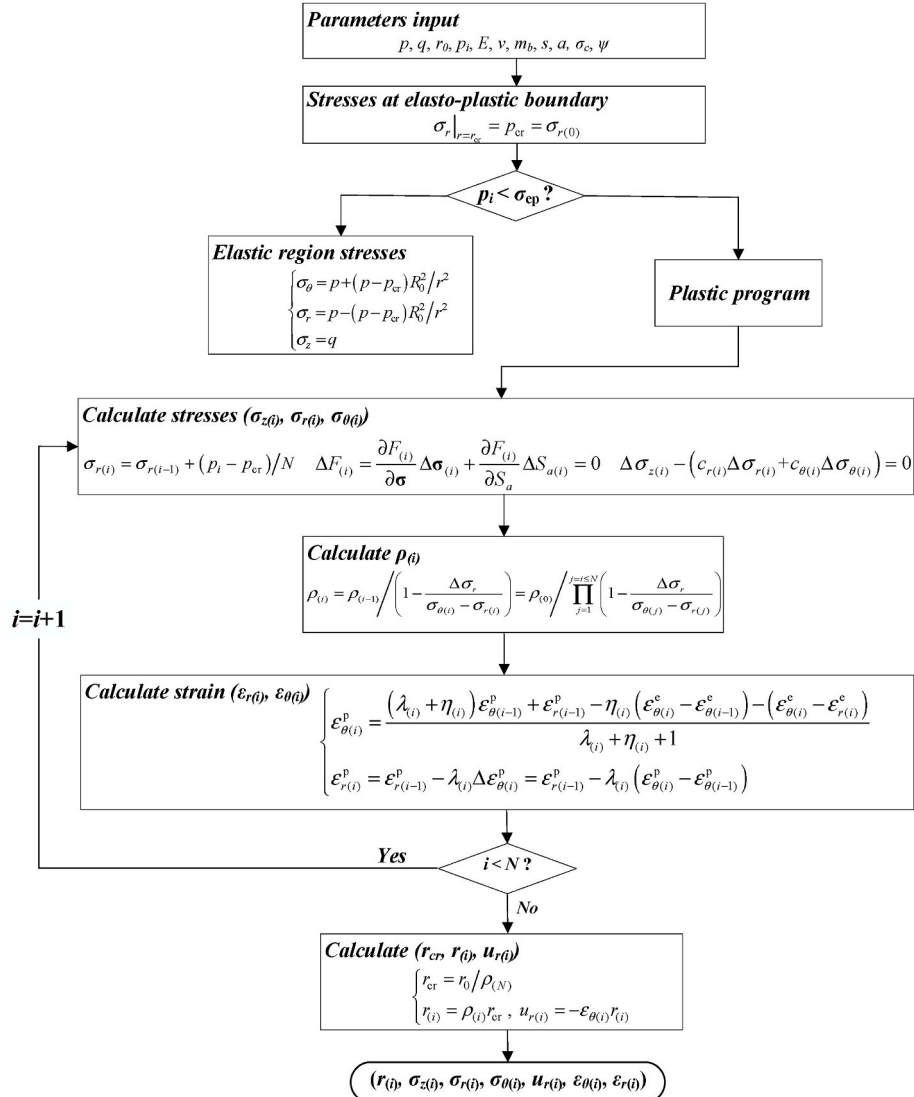


Fig. 3. Flow chart for solving semi-analytical elasto-plastic response presented in this study.

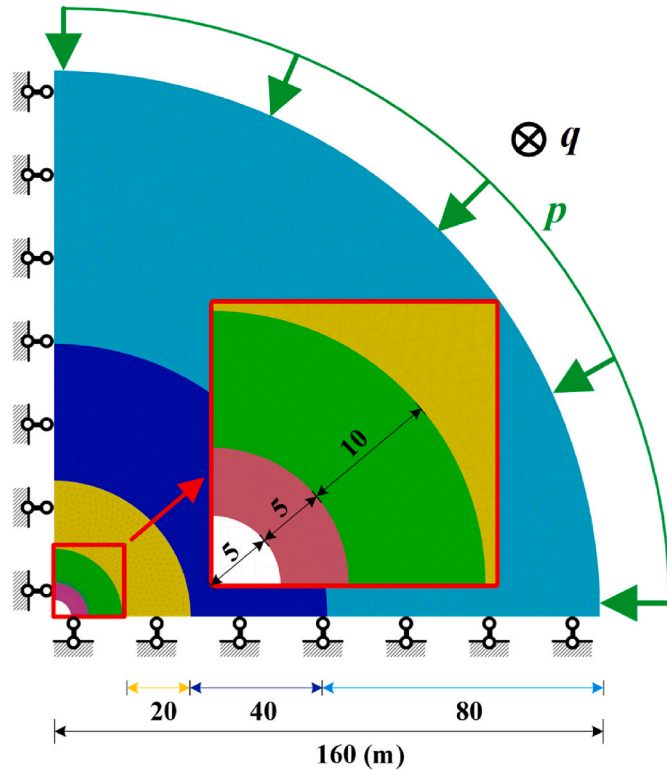


Fig. 4. FEM modeling of one-quarter thick-walled cylinder.

Table 2
Parameters of surrounding rock mass.

r_0	p_0	σ_{ci}	GSI	E	ν	m_i	p_i	Ψ
5 m	40 MPa	50 MPa	50	2.0 GPa	0.35	6	0.5 MPa	8°

$$\Delta\sigma_{z(i)} - (c_{r(i)}\Delta\sigma_{r(i)} + c_{\theta(i)}\Delta\sigma_{\theta(i)}) = 0 \quad (49c)$$

The incremental stress of Eq. (49) is

$$\Delta\sigma_{\theta(i)} = -\frac{(L_{r(i)} + c_{r(i)}L_{z(i)})\Delta\sigma_r + (\partial F_{(i)}/\partial S_a)\Delta S_{a(i)}}{(L_{\theta(i)} + c_{\theta(i)}L_{z(i)})} \quad (50a)$$

$$\Delta\sigma_{z(i)} = \frac{(L_{\theta(i)}c_{r(i)} - L_{r(i)}c_{\theta(i)})\Delta\sigma_r - (\partial F_{(i)}/\partial S_a)\Delta S_{a(i)}}{(L_{\theta(i)} + c_{\theta(i)}L_{z(i)})} \quad (50b)$$

where $L_{r(i)} = \frac{\partial F_{(i)}}{\partial \sigma_r}$, $L_{\theta(i)} = \frac{\partial F_{(i)}}{\partial \sigma_\theta}$, and $L_{z(i)} = \frac{\partial F_{(i)}}{\partial \sigma_z}$. If the associated flow rule is considered ($F = Q$) in the plane strain model, we can obtain $d\epsilon_z^p = d\lambda \frac{\partial F_{(i)}}{\partial \sigma_z} = d\lambda L_{z(i)} = 0$. If the elastic-plastic constitutive model is used, Eq. (50) can be transformed into

$$\Delta\sigma_{\theta(i)} = -\frac{L_{r(i)}\Delta\sigma_r}{L_{\theta(i)}} \quad (51a)$$

$$\Delta\sigma_{z(i)} = \frac{L_{\theta(i)}c_{r(i)} - L_{r(i)}c_{\theta(i)}}{L_{\theta(i)}}\Delta\sigma_r \quad (51b)$$

According to the theory of elastic mechanics, the elastic stress is independent of the rock mass strength and deformation parameters. However, it can be inferred from Eq. (50) that the stress in the plastic region is closely related to the stress state, rock mass strength, and deformation parameters, which greatly increases the difficulties in the semi-analytic solution for plastic analysis. The iterative solution for nonlinear full equations (i.e., $F_{(i)} = 0$ and $G_{(i)} = 0$) is very tedious and

time consuming, but the efficiency of the solution can be significantly improved if the closed-form analytical incremental solution (i.e., Eq. (50) or (51) is used for the calculation.

3.2.5. Strain and displacement in the plastic zone

The stress components in the plastic zone satisfy the equilibrium equation

$$\frac{d\sigma_r}{dr} - \frac{\sigma_\theta - \sigma_r}{r} + F_r = 0 \quad (52)$$

Here, F_r is the volume force in the radial direction, and $F_r = 0$ is often assumed in the numerical solution. The difference form of Eq. (52) can be written as

$$\frac{\Delta\sigma_r}{\rho_{(i)} - \rho_{(i-1)}} - \frac{\sigma_{\theta(i)} - \sigma_{r(i)}}{\rho_{(i)}} = 0 \quad (53)$$

The solution of Eq. (53) is

$$\rho_{(i)} = \rho_{(i-1)} \left/ \left(1 - \frac{\Delta\sigma_r}{\sigma_{\theta(i)} - \sigma_{r(i)}} \right) \right. = \rho_{(0)} \left/ \prod_{j=1}^{i-1} \left(1 - \frac{\Delta\sigma_r}{\sigma_{\theta(j)} - \sigma_{r(j)}} \right) \right. \quad (54)$$

$\rho_{(i)}$ can be obtained from Eq. (54).

The strain component of the i -th ring can be decomposed into

$$\begin{Bmatrix} \epsilon_{r(i)} \\ \epsilon_{\theta(i)} \end{Bmatrix} = \begin{Bmatrix} \epsilon_{r(i-1)} \\ \epsilon_{\theta(i-1)} \end{Bmatrix} + \begin{Bmatrix} \Delta\epsilon_{r(i)}^e \\ \Delta\epsilon_{\theta(i)}^e \end{Bmatrix} + \begin{Bmatrix} \Delta\epsilon_{r(i)}^p \\ \Delta\epsilon_{\theta(i)}^p \end{Bmatrix} \quad (55)$$

where $\Delta\epsilon^e$ and $\Delta\epsilon^p$ denote the elastic and plastic deformation increments, respectively, and $\Delta\epsilon^e$ can be obtained from the incremental Hooke's law (Eq. (31)).

The strain satisfies the compatibility equation⁵⁶

$$\frac{d\epsilon_\theta}{dr} + \frac{\epsilon_\theta - \epsilon_r}{r} = 0 \quad (56)$$

Expressing the strain component as the sum of the elastic strain and plastic strain, then, Eq. (56) can be written as

$$\frac{d\epsilon_\theta^p}{dr} + \frac{\epsilon_\theta^p - \epsilon_r^p}{r} = -\frac{d\epsilon_\theta^e}{dr} - \frac{\epsilon_\theta^e - \epsilon_r^e}{r} \quad (57)$$

The difference form of Eq. (57) can be written as

$$\frac{\epsilon_{\theta(i)}^p - \epsilon_{\theta(i-1)}^p}{\rho_{(i)} - \rho_{(i-1)}} + \frac{\epsilon_{\theta(i)}^p - \epsilon_{r(i)}^p}{\rho_{(i)}} = -\frac{\epsilon_{\theta(i)}^e - \epsilon_{\theta(i-1)}^e}{\rho_{(i)} - \rho_{(i-1)}} - \frac{\epsilon_{\theta(i)}^e - \epsilon_{r(i)}^e}{\rho_{(i)}} \quad (58)$$

Considering $d\epsilon_z^p = 0$ in the plane strain model, $d\epsilon_r^p$ and $d\epsilon_\theta^p$ satisfy

$$d\epsilon_r^p + \lambda d\epsilon_\theta^p = 0 \quad (59)$$

Its difference form is given as

$$\Delta\epsilon_{r(i)}^p + \lambda_{(i)}\Delta\epsilon_{\theta(i)}^p = 0 \quad (60)$$

The MC-based non-associated flow coefficient ($\lambda_{(i)} = \frac{1 + \sin \psi_{(i)}}{1 - \sin \psi_{(i)}}$) has often been used in previous studies. In this study, $\lambda_{(i)}$ can be obtained through the plastic mechanics method using Eq. (38)

$$\lambda_{(i)} = -\frac{d\epsilon_{r(i)}^p}{d\epsilon_{\theta(i)}^p} = -\frac{\frac{1}{3} + \frac{g(J_{2(i)}, \theta_{\theta(i)})}{\sqrt{J_{2(i)}}}S_{r(i)} - \frac{3}{2J_{2(i)}}S_{r(i)}^2}{\frac{1}{3} + \frac{g(J_{2(i)}, \theta_{\theta(i)})}{\sqrt{J_{2(i)}}}S_{\theta(i)} - \frac{3}{2J_{2(i)}}S_{\theta(i)}^2} \quad (61)$$

An iterative solution is not required because the stress components in Eq. (61) are known for the i -th circular ring. It should be noted that unlike in previous studies, $\epsilon_{r(i)}^p + \lambda_{(i)}\epsilon_{\theta(i)}^p = 0$ cannot be deduced from Eq. (59) because of the nonconstant value of $\lambda_{(i)}$. Therefore, the term $\epsilon_{\theta(i)}^p - \epsilon_{r(i)}^p$ is transformed using Eq. (60) to obtain

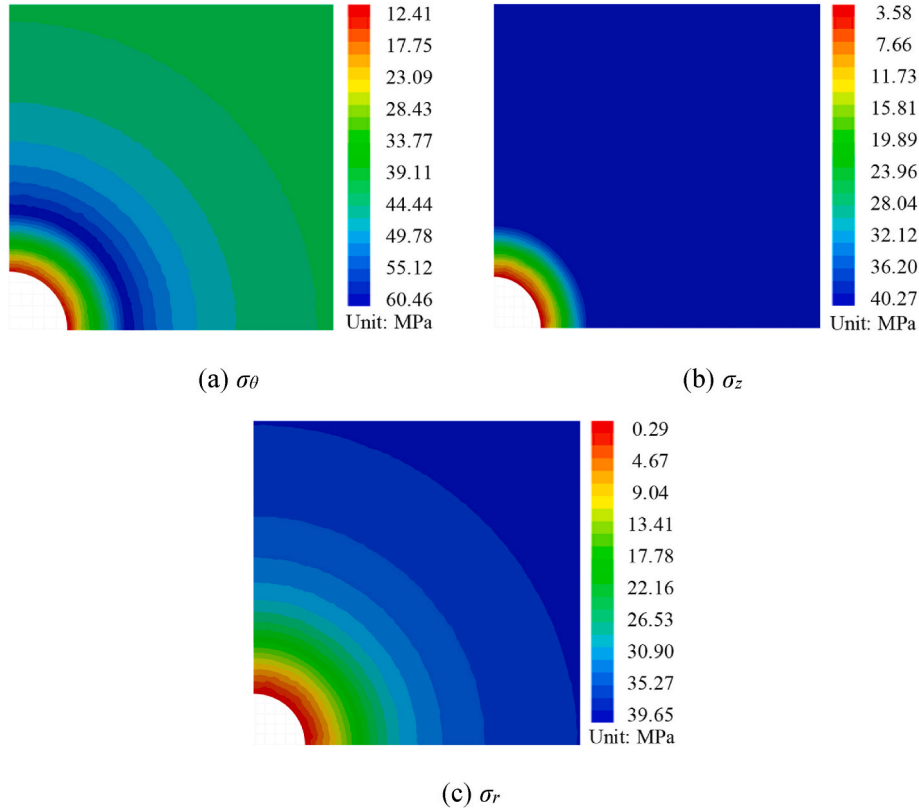


Fig. 5. Numerical results of stress cloud.

$$\begin{aligned} \epsilon_{\theta(i)}^p - \epsilon_{r(i)}^p &= (\epsilon_{\theta(i-1)}^p + \Delta \epsilon_{\theta(i)}^p) - (\epsilon_{r(i-1)}^p + \Delta \epsilon_{r(i)}^p) = (\epsilon_{\theta(i-1)}^p - \epsilon_{r(i-1)}^p) \\ &+ (1 + \lambda_{(i)}) \Delta \epsilon_{\theta(i)}^p \end{aligned} \quad (62)$$

Substituting Eq. (62) into (58), we obtain

$$\epsilon_{\theta(i)}^p = \frac{(\lambda_{(i)} + \eta_{(i)}) \epsilon_{\theta(i-1)}^p + \epsilon_{r(i-1)}^p - \eta_{(i)} (\epsilon_{\theta(i)}^c - \epsilon_{\theta(i-1)}^c) - (\epsilon_{\theta(i)}^c - \epsilon_{r(i)}^c)}{\lambda_{(i)} + \eta_{(i)} + 1} \quad (63a)$$

$$\epsilon_{r(i)}^p = \epsilon_{r(i-1)}^p - \lambda_{(i)} \Delta \epsilon_{\theta(i)}^p = \epsilon_{r(i-1)}^p - \lambda_{(i)} (\epsilon_{\theta(i)}^p - \epsilon_{\theta(i-1)}^p) \quad (63b)$$

where $\eta_{(i)} = \frac{1}{1 - \frac{\rho_{(i-1)}}{\rho_{(i)}}}$.

$\epsilon_{\theta(i)}$ and $\epsilon_{r(i)}$ can be obtained by superimposing the corresponding elastic and plastic stress components, and the radial displacement (u_r) can be solved using $u_{r(i)} = -\epsilon_{\theta(i)} r_{(i)}$. The $\rho(n)$ and the $r_{cr} = \frac{r_0}{\rho_{(n)}}$ can be obtained when $i = n$, and hence, the radius of the i -th circular ring can be obtained using $r_{(i)} = r_{cr} \rho_{(i)}$.

The flow chart for solving elasto-plastic response is shown in Fig. 3.

4. Numerical verification of different out-of-plane stress equations

An FEM numerical model for a plane strain tunnel was established in this study. The numerical results were compared with those obtained from the different out-of-plane stress equations presented in Sections 3.2.3 and 3.2.4. As shown in Fig. 4, one-quarter thick-walled cylinder numerical model was established, the rock mass was divided into five rings with radii of 5 m, 10 m, 20 m, 40 m, and 80 m. The vertical fixed constraints were applied on the left, right, and back sides of the model to reduce the degree of freedom of the rock mass and increase the stability of the numerical calculation. The rock mass parameters and excavation

information are shown in Table 2.

Smooth GZZ-based numerical program was developed to facilitate comparison with the analytical method used in this study. The advantages and limitations of different approaches were analyzed, and the effect of the out-of-plane stress on the strength, deformation, and stability of the surrounding rock under hydrostatic stress conditions (i.e., $q = p = 40$ MPa) were analyzed using different out-of-plane stress equations. The numerical results are shown in Fig. 5, and the corresponding comparison of the semi-analytical and numerical results is shown in Fig. 6.

In general, the rock mass strengths at the elastic-plastic boundary obtained using different equations agree well with the numerical results, but there are obvious differences in the stress results. Fig. 6(a) and (c) show that there is an obvious stress jump at the elastic-plastic boundary for $G_{(i)}^I = 0$ and $G_{(i)}^{III} = 0$ because the stress at the elastic-plastic boundary does not satisfy the supplementary equation in the plastic zone. Although the stress continuity condition can be satisfied for $G_{(i)}^{II} = 0$ (Fig. 6(b)), r_{cr} and stress are significantly different from those of the numerical solution, and there is a stress crossover near the tunnel wall (stress state translated from $\sigma_{\theta} > \sigma_z$ to $\sigma_{\theta} < \sigma_z$). The main reason is that there is no exact solution near the tunnel wall using an elastic approach, and the approximate solution has a significant effect on the dimensionless radius $\rho_{(n)}$, resulting in a large error in the radius and stress distribution in the plastic zone.

The stress drops in the $G_{(i)}^{III} = 0$ approach cause the σ_z in the plastic zone to be higher than that of numerical results, and this is the main reason for the difference between the semi-analytical and numerical solutions when calculating σ_r and σ_{θ} . The semi-analytical results based on the plastic approach ($G_{(i)}^{IV} = 0$) agree well with the numerical solution, but r_{cr} is slightly smaller than that of the numerical solution, and σ_z near r_{cr} is slightly larger than that of the numerical solution. The overall

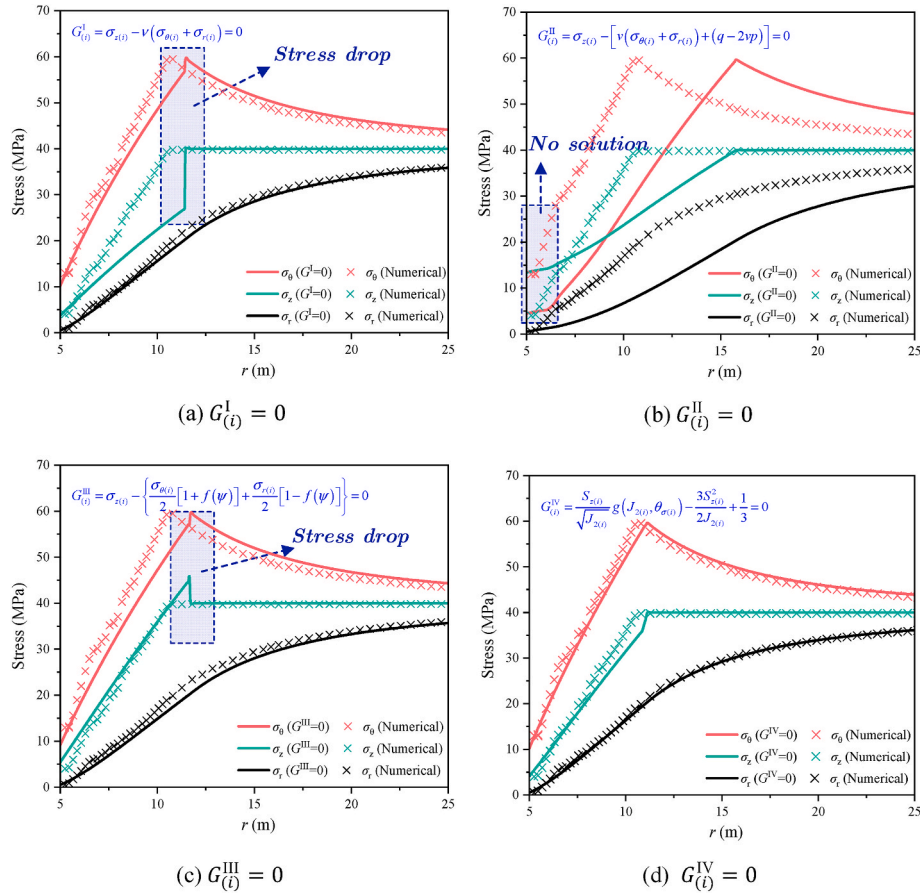


Fig. 6. Numerical and semi-analytical solutions obtained from different out-of-plane stress equations.

accuracy of the plastic approach is far higher than those of the other three approaches, and particularly higher than those of the elastic approaches.

Among the four approaches, the σ_z in the plastic zone based on the elastic approach ($G_{(i)}^I = 0$ and $G_{(i)}^{II} = 0$) is significantly different from those of the numerical results, indicating that the elastic Hooke's law is not suitable as the supplementary equation in stress calculation. The parameter $c_r = c_\theta = \nu$ is set in the elastic approach, suggesting that the contributions of σ_r and σ_θ to σ_z are the same and linear. However, the dilatancy-dependent and plastic mechanics approaches ($G_{(i)}^{III} = 0$ and $G_{(i)}^{IV} = 0$) have high accuracies, and both approaches consider $c_r \neq c_\theta$, reflecting the phenomenon of “separation of Poisson's ratio”, which suggests that the contributions of σ_r and σ_θ to σ_z are different and non-linear.

$G_{(i)}^{III} = 0$ is an empirical equation based on the FEM results reported by Pan and Brown,⁴⁰ the result of which is similar to that of $G_{(i)}^{IV} = 0$ based on plastic theory and $d\epsilon_z^p = 0$. However, $G_{(i)}^{III} = 0$ is mainly used for the hydrostatic stress condition and plastic area near the tunnel wall.⁴⁰ $G_{(i)}^{III} = 0$ has poor applicability under non-hydrostatic stress condition, as shown in Fig. 7(a) and (b), and σ_z exhibits significant stress jump at the elastic–plastic boundary, which is significantly different from that of the numerical solution, leading to calculation error of the rock strength and stress.

For the stress calculated using $G_{(i)}^{IV} = 0$ and $q = 20$ MPa, the semi-analytical solution agrees well with the numerical result, as shown in Fig. 7(c). The σ_z of the semi-analytical results based on $G_{(i)}^{IV} = 0$ obtain relatively higher accuracy over other approaches, and exhibit a similar trend with numerical results. But there are still slight differences near

the elastic–plastic boundary, which causes a slight deviation between the semi-analytical results and numerical results for σ_r and σ_θ . For $q = 60$ MPa in Fig. 7(d), there is no exact solution near the elastic–plastic boundary because of the nonlinear characteristics of the plastic approach. Therefore, the deviations between the semi-analytical results and numerical results gradually increase. The deviation gradually decreases when moving away from the solution-free region (or near the elastic–plastic boundary region). Therefore, the calculation error of the overall plastic stress is caused by the impracticability of the large value of q , which also demonstrates the importance of choosing a reasonable out-of-plane stress equation.

It can be observed from the analytical and numerical calculation results that σ_z first increases and then decreases when $q < p$, and gradually decreases when $q > p$. It should be noted that the σ_z near the tunnel wall softens to σ_2 regardless of the value of q . It can be observed that the out-of-plane and in-plane stresses are closely related in the plastic zone, and the accuracy of the 3D semi-analytical approach exhibits significant σ_z -dependence, and σ_z directly affects the rock mass strength, radius of the plastic zone, stress state in the plastic zone, and stability of the surrounding rock mass.

As shown in Fig. 8(a) for the elastic approach ($G_{(i)}^{II} = 0$), σ_z remains constant in the elastic region and linearly softens in the plastic region, which shows a significant dependence on q because q is included in the equation for $G_{(i)}^{II} = 0$. The softening rates under different q values are completely different, and the plastic radius is also significantly different, exhibiting an obvious abnormal inflection point for $q = 45$ MPa and 50 MPa near the tunnel wall when $G_{(i)}^{II} = 0$. However, for the plastic approach ($G_{(i)}^{IV} = 0$), σ_z in the plastic region consists of a quick adjustment zone and a uniform softening zone, as demonstrated by the softening feature of the curve trend in Fig. 8(b). In the quick adjustment

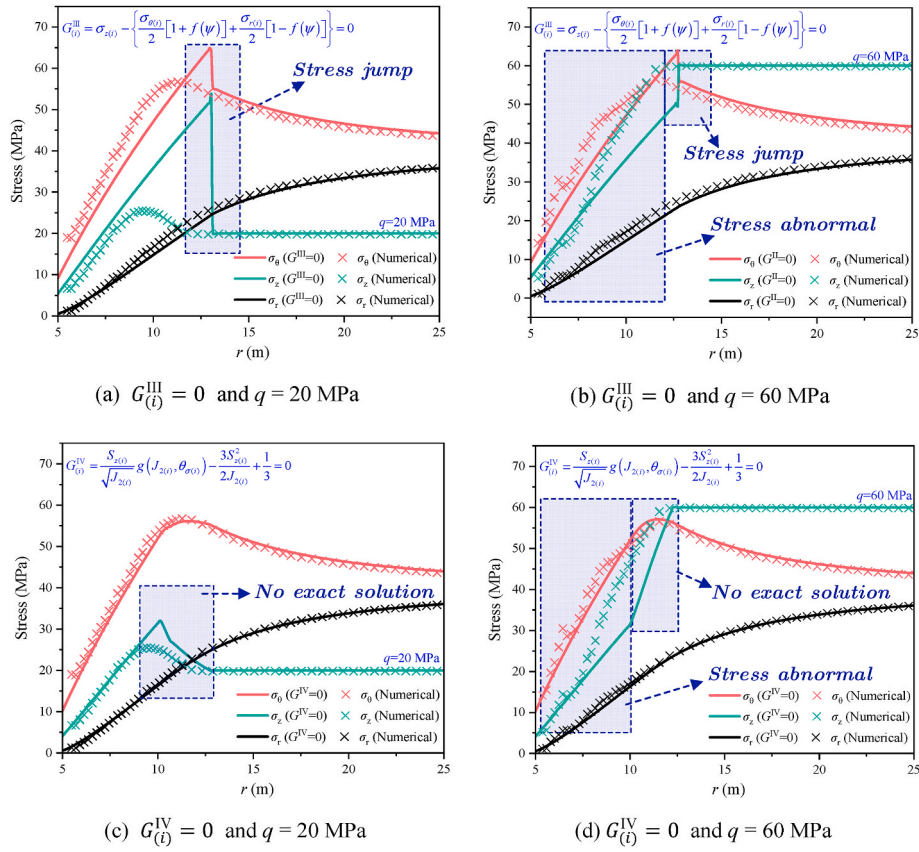


Fig. 7. Calculation results of $G_{(i)}^{III} = 0$ and $G_{(i)}^{IV} = 0$ at $q = 20$ MPa and $q = 60$ MPa.

zone, σ_z first increases and then decreases from the depth of the surrounding rock to the tunnel wall at a smaller q value (i.e., $q = 30$ MPa and 35 MPa in Fig. 8(b)). At a larger q value (i.e., $q = 40$ MPa, 45 MPa, and 50 MPa in Fig. 8(b)), the σ_z values obtained from the numerical results exhibit almost linear softening law, whereas the analytical solution ($G_{(i)}^{IV} = 0$) first shows a rapid decrease and then enters the uniform softening zone. The σ_z values in the uniform softening zone are almost the same at different q values, and the numerical solution and analytical solution are in good agreement.

There are differences in the σ_z values at different q values in the quick adjustment zone using $G_{(i)}^{IV} = 0$, which leads to differences in the in-plane stress (σ_r and σ_θ) in the quick adjustment zone at different q values, and hence, differences in the deformation. The rapid softening σ_z or stress release in the quick adjustment zone causes radial displacement (u_r), but difference in u_r obtained by different q is limited because of the small range of the quick adjustment zone. It can be observed that the out-of-plane stress closely correlates with the in-plane stress, and this correlation may be affected by q in the quick adjustment zone, but hardly affected by q in the uniform softening zone.

5. Case analysis

The analysis in Section 4 shows that the plastic approach ($G_{(i)}^{IV} = 0$) generally gave the best performance among the four approaches. Elastic approach (i.e., $G_{(i)}^{II} = 0$) based results do not satisfy the equation $G_{(i)}^{IV} = 0$, and we can obtain $G_{(i)}^{IV} \neq G_{(i-1)}^{IV}$ and $d\epsilon_{z(i)}^p \neq d\epsilon_{z(i-1)}^p \neq 0$, which may result in strain distortion for different circular rings. This implies that the stress and strain obtained using elastic approaches may not be the true plane strain tunnel model.

In this section, $G_{(i)}^{IV} = 0$ will be used for case analysis considering strain softening performance to realistically reflect the mechanical

properties of the engineering rock mass. The softening law of the strength parameters can be obtained by substituting $GSI(\gamma^p)$ in Eq. (8) into Eq. (2), and the deformation parameters can be obtained from Eq. (10). The calculation parameters are $r_0 = 5$ m, $GSI = 50$, $\sigma_{ci} = 50$ MPa, $E = 4.5$ GPa, $\nu = 0.35$, $m_i = 6$, $D = 0$, $p_i = 2$ MPa, and $\varphi = 50^\circ$.

5.1. Influence of GSI on the surrounding rock stress and deformation

$GSI = 30$ is considered to be poor quality rock mass with a well-developed structural plane, whereas $GSI = 60$ is considered to be good quality rock mass with a slightly-developed structural plane. Fig. 9 shows the distribution and softening laws of σ_z for different GSIs. In general, the distribution of the σ_z at different q values is consistent with that presented in Section 4.2. σ_z in the plastic region can be divided into a quick adjustment zone and a uniform softening zone, and the σ_z values in the uniform softening zone exhibit almost the same trend at different q values.

The surrounding rock deformation mainly occurs in the uniform softening zone for the strain-softening rock mass, and the smaller the range of the quick adjustment zone, the smaller the difference between the analytical result obtained using $G_{(i)}^{IV} = 0$ and the numerical solution. The red shaded area in Fig. 9 is the quick adjustment zone obtained at $q = 40$ MPa. It can be observed that the larger the GSI, the smaller the range of the quick adjustment zone and the smaller the error. Therefore, $G_{(i)}^{IV} = 0$ is more applicable to high quality rock mass.

There are large differences in the distribution of the plastic zone under different GSIs, and a smaller GSI corresponds to a larger plastic radius. In addition, a smaller GSI corresponds to a larger difference in the plastic radius for different q values, which ensures that the sensitivity of the plastic radius (or deformation) to q is closely related to the quality of the rock mass, and the poor quality of the rock mass corresponds to the greater effect of q on the plastic radius and deformation.

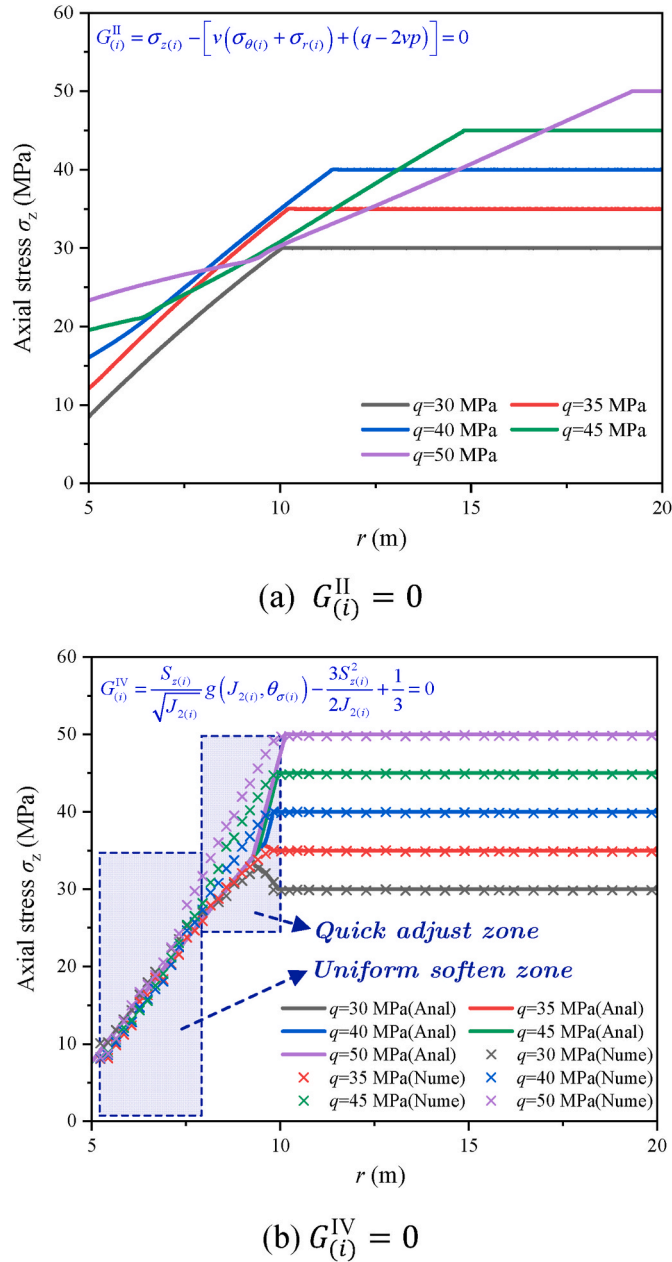


Fig. 8. Distributions of σ_z for $G_{(i)}^{II} = 0$ and $G_{(i)}^{IV} = 0$ for different q values ($p_1 = 2$ MPa).

5.2. Effect of q on the surrounding rock stress and deformation

Many previous studies have reported that the out-of-plane stress has a significant influence on the in-plane stress and the overall stability of the surrounding rock^{27,39–41,57} through analytical observations, which can be illustrated in Fig. 10(a) (elastic approach). Different q values correspond to different tunnel wall deformations, and both approaches ($G_{(i)}^{II} = 0$ and $G_{(i)}^{IV} = 0$) assume that there is an optimal q at which the deformation of the surrounding rock is minimized and the plastic radius is the smallest. Pan et al.⁴⁰ also noted that the surrounding rock exhibited better stability when $p = q$. In this study, the optimal in-situ out-of-plane stresses (q_{opt}) based on elastic and plastic approaches are 31 MPa and 38 MPa, respectively, and the plastic approach-based results are slightly smaller than p , demonstrating good consistency with numerical results and the conclusions made by Pan et al.⁴⁰ When selecting the tunnel site, the tunnel surveyor and designer can select an optimal

tunnel axial direction in line with q_{opt} to ensure tunnel stability.

However, there is a significant difference between the results obtained for $G_{(i)}^{II} = 0$ and $G_{(i)}^{IV} = 0$. The deformations of the surrounding rock obtained using $G_{(i)}^{II} = 0$ for different q values differ by up to a factor of ten, whereas the effect of q on the surrounding rock deformation and stability is limited when $G_{(i)}^{IV} = 0$ is used in the semi-analytical research, and the differences of the plastic radius and deformations for different q values are generally less than 10%. Therefore, the effect of the in-situ out-of-plane stress will be exaggerated if unsuitable softening equations are used (e.g., generalized Hooke's law), leading to some incorrect conclusions.

5.3. Surrounding rock stresses and deformations for different constitutive models

Fig. 11(a) to (c) show plots of σ_θ , σ_z , and u_r using e-b-p ($GSI_p = 50$ and $GSI_r = 25$), e-p-p ($GSI_p = GSI_r = 50$), and e-s-s ($GSI_p = 50$ and $GSI_r = 45$, 40, 35, 30, and 25) constitutive models. The rock strengths or the peak σ_θ of different constitutive models are the same because the GSI_p values for different constitutive models are equal.

The deformation and plastic radius using the e-p-p model are the smallest, as the GSI of this model is the largest. Similarly, the deformation and plastic radius using the e-b-p model are the largest as the GSI of this model is always residual when entering the plastic zone, and there is an obvious stress drop at the elastic-plastic boundary. For the e-s-s model, the plastic radius and deformation exhibit obvious nonlinear characteristics with a decrease in GSI_r , and a smaller GSI_r corresponds to a higher sensitivity of the surrounding rock to the change in GSI_r .

5.4. Effect of rock mass dilatancy on the surrounding rock stress and stability

The rock mass produces obvious volume dilatancy under excavation unloading, particularly for deep tunnels under high ground stresses. The dilatancy effect of rock mass is similar to Poisson effect, and both are important physical phenomena that reflect the expansion characteristics of rock masses. The dilatancy angles for the rock masses of different qualities were varied; for instance, $\psi = \frac{\pi}{4}$ is for high quality rock mass, $\psi = \frac{\pi}{8}$ is for average quality rock mass, and $\psi = 0$ is for poor quality rock mass based on Hoek's recommendation.⁴⁹ The effect of rock mass dilatancy on the surrounding rock stress seems insignificant, as shown in Fig. 12, and different curves of the rock almost overlap at varying dilatancy angles. The effect of dilatancy on deformation is very significant, and the deformation exhibits a significant nonlinear trend with the increase in the dilatancy angle.

5.5. Mechanical response of surrounding rock based on HB and GZZ

According to the conclusion in Section 4.2, σ_z will uniformly soften to σ_2 in the plastic zone near the tunnel wall at different q values, and analysis of σ_z on the surrounding rock stability focused on the effect of σ_2 in practice. Fig. 13 shows the mechanical responses of the tunnel excavation obtained using HB and GZZ. The rock mass strength (σ_θ) at the elastic-plastic boundary based on HB is slightly smaller than that obtained using GZZ, and the plastic radius and stress distributions obtained using the two criteria are significantly different. Meanwhile, the differences in the strength and stress of the two criteria will further lead to a difference in the surrounding rock deformation, and the deformation obtained using HB is 2 ~ 3 times that of GZZ. This suggests that conservative results are obtained using HB, which will lead to a significant error when analyzing deep tunnel stability, and underestimating the rock mass strength will lead to waste of support materials and cause "transitional support".

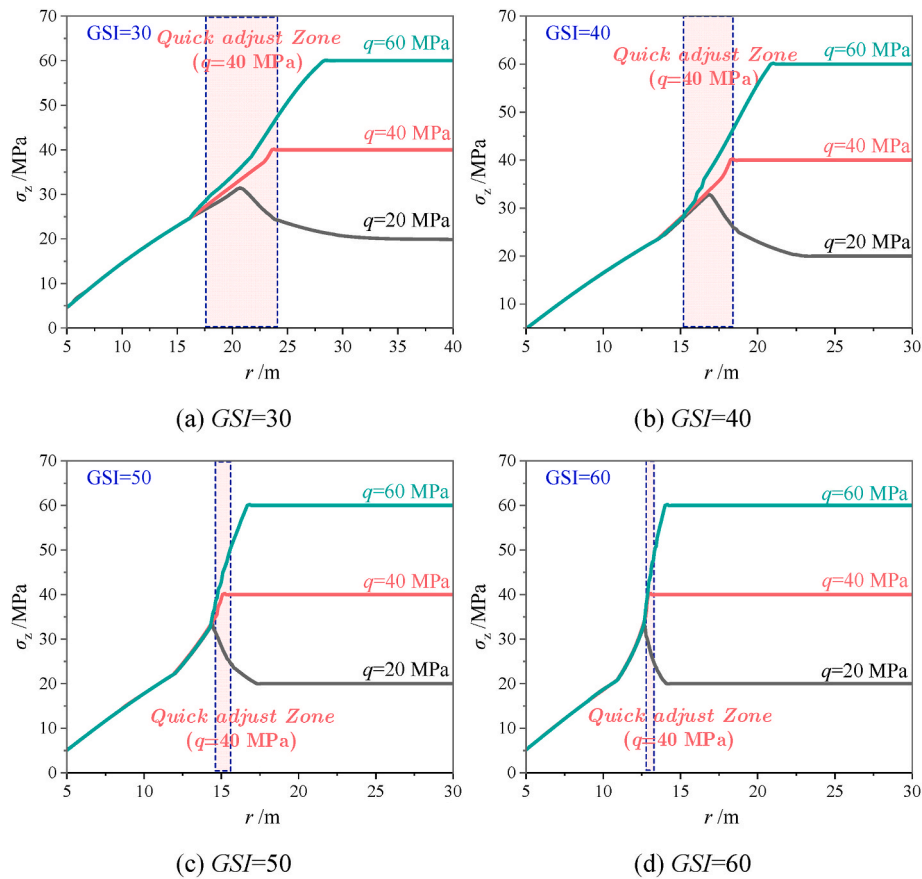


Fig. 9. Influence of GSI on σ_z at different q values.

6. Engineering application and discussion

The mechanical properties of rock mass directly affect the safety and stability of tunnel engineering,^{58–64} and this part mainly explores its influence on support design. GRC can reflect the nonlinear mechanical behaviors of the surrounding rock, and combining it with the support characteristic curve (SCC) is an important basis for support measures to ensure rock stability. Different GSI values correspond to different GRC curves, as shown in Fig. 14. A smaller GSI (poor quality rock mass) corresponds to a larger convergence deformation, and a higher support stiffness is required to control the deformation to the same standard.

The GRCs obtained using HB and GZZ exhibit significant differences, and they also require different support stiffnesses to control deformation. For instance, for the HB-based GRC (Fig. 14(a)), a support stiffness of $\frac{\Delta p_i}{\Delta u^*} = 1.85$ is required for the surrounding rock ($GSI = 30$) to be controlled to $0.12u^*$, whereas for the GZZ-based GRC (Fig. 14(b)), a support stiffness of $\frac{\Delta p_i}{\Delta u^*} = 1.11$ is sufficient to achieve the same control goal. It can be observed that the required stiffness using HB is 1.67 times that of GZZ. For high quality rock mass ($GSI = 60$), the support stiffness using HB and GZZ are 0.88 and 0.27, respectively, indicating that the required stiffness using HB is 3.26 times that of GZZ. Therefore, the support measures based on 2D and 3D strength criteria exhibited relatively significant differences, particularly for rock mass with high quality.

On the other hand, the HB-based support stiffness for rock mass with $GSI = 45$ is 1.25, and the GZZ-based support stiffness for rock mass with $GSI = 60$ is 1.11, demonstrating that using an inappropriate strength criterion (e.g., 2D HB) has a significant effect on proper selection of support measures, and its effect can even exceed the effect of misjudging the rock mass quality to a certain extent.

Similar to the analysis method of the strength criterion, the

supplementary equation (the relationship between the out-of-plane stress and in-plane stress) will also affect the stability analysis of the surrounding rock and preliminary design of support measures. In this study, σ_z quickly softens to σ_2 in the plastic zone near the tunnel wall under different q values. However, some divergences exist in the conclusions of other analytical approaches. For instance, stress crossing exists between σ_z and σ_θ in the plastic zone based on the elastic approach, as shown in Fig. 6(b), which is very different from that in the numerical solution. The main reason is that unreasonable elastic theories were used for the semi-analytical solutions, and this elastic approach (e.g., generalized Hooke's law) exaggerates the influence of q to a certain extent, leading to wrong conclusions.

For deep rock mechanics and tunnel engineering, the plastic zone of tunnel excavation is widely distributed. The applicability of elastic theories and engineering design experiences gathered from shallow tunnels should be further considered, and this has an important effect on the stability analysis of deep rock masses and reasonable selection of support measures.

7. Conclusions

The semi-analytical solution of a circular tunnel excavation was deduced and developed based on the smooth GZZ strength criterion and four different out-of-plane stress equations, and the applicability of different approaches was compared and analyzed using the numerical results. The following conclusions are drawn based on the findings of this study.

- (1) It is difficult to find a universal out-of-plane stress equation for analytical or semi-analytical solution. Some differences exist in the mechanical responses of the surrounding rock based on the

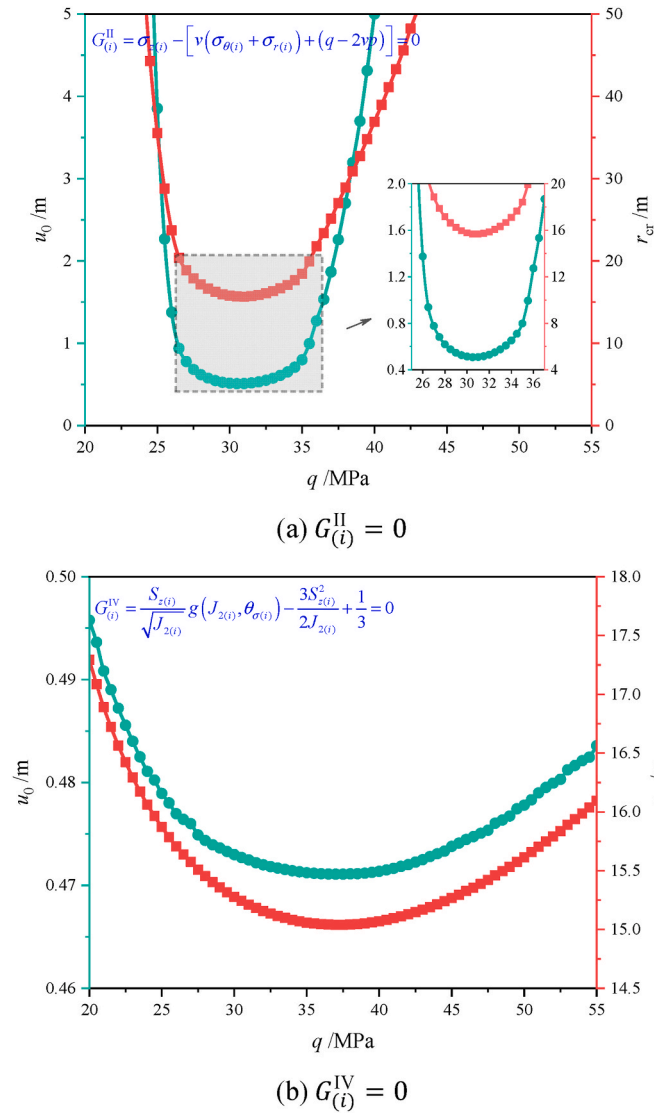


Fig. 10. Variations of u_0 and r_{cr} with q for $G_{(i)}^H = 0$ and $G_{(i)}^{IV} = 0$ ($p_i = 2$ MPa).

- different out-of-plane stress equations. The different equations can be represented as a unified incremental equation, i.e., $\Delta\sigma_{z(i)} = (c_{r(i)}\Delta\sigma_{r(i)} + c_{\theta(i)}\Delta\sigma_{\theta(i)})$. Elastic approaches (e.g., Hooke's law) consider that $c_r = c_{\theta} = \nu$ and the contributions of σ_r and σ_{θ} to σ_z are the same and linear, which is not applicable to the plastic zone. However, plastic approaches-based results agree well with numerical results and have high accuracies as they consider that $c_r \neq c_{\theta}$, suggesting that the contributions of σ_r and σ_{θ} to σ_z are different and non-linear.
- (2) The plastic zone consists of quick adjustment zone and uniform softening zone according to the softening law of σ_z . The σ_z in the uniform zone shows good consistency even under different in-situ out-of-plane stresses (q), indicating a close correlation between the out-of-plane stress and in-plane stress. There is a significant difference between the results obtained using elastic approach and plastic approach. Elastic approach-based excavation response exhibits a significant dependence on the q ; but the influence of q is limited when the smooth GZZ-based plastic approach is used in the semi-analytical research. So, the effect of q is exaggerated if inappropriate softening equations (e.g., the generalized Hooke's law) are used, leading to incorrect conclusions.

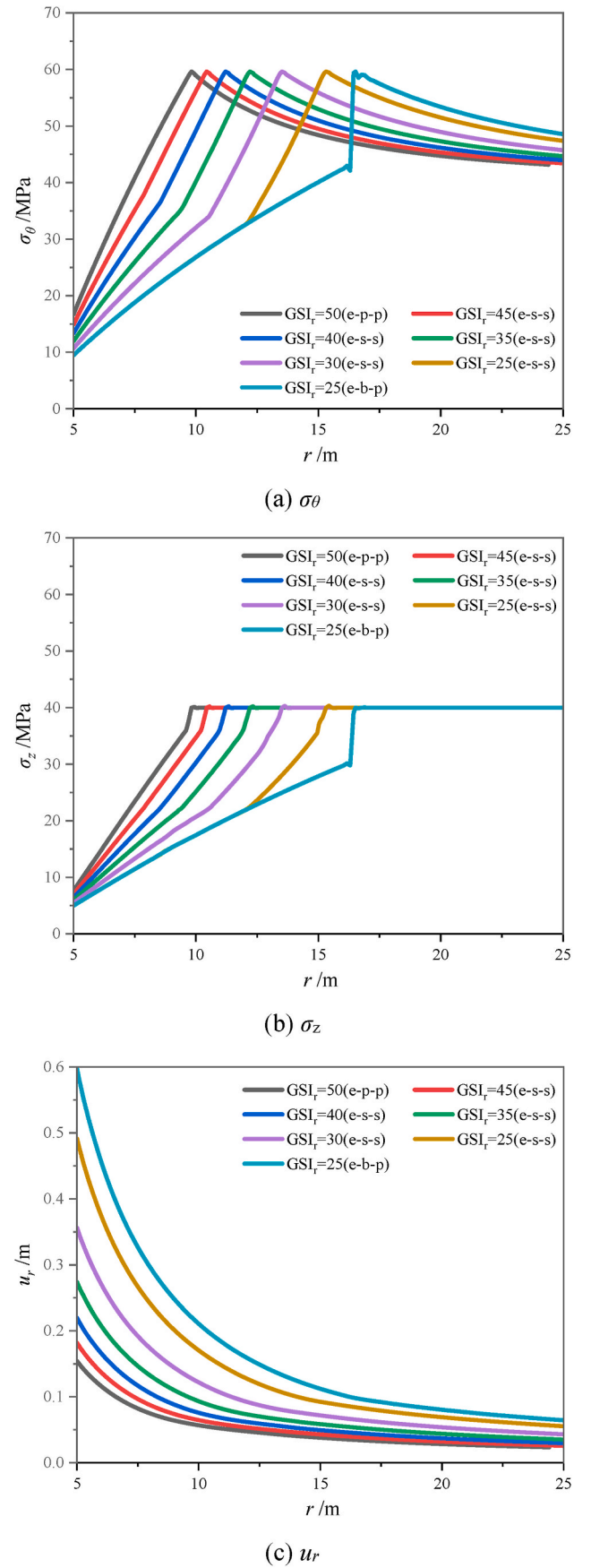


Fig. 11. Mechanical responses of the surrounding rock for different constitutive models ($p = q = 40$ MPa).

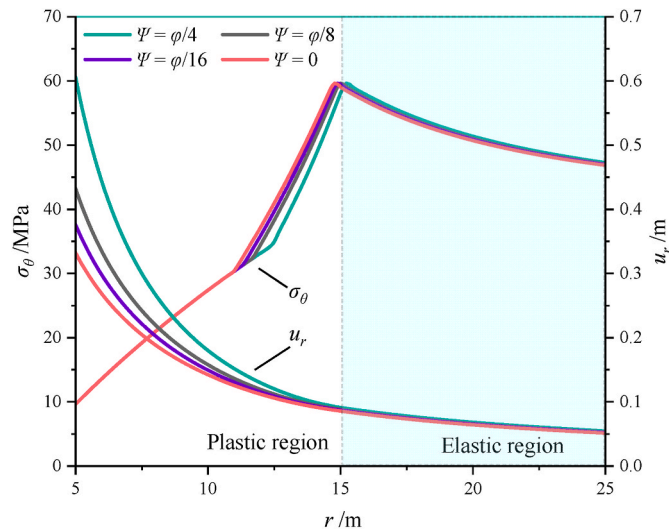


Fig. 12. Mechanical responses of surrounding rock at varying dilatancy angles ($p = q = 40$ MPa).

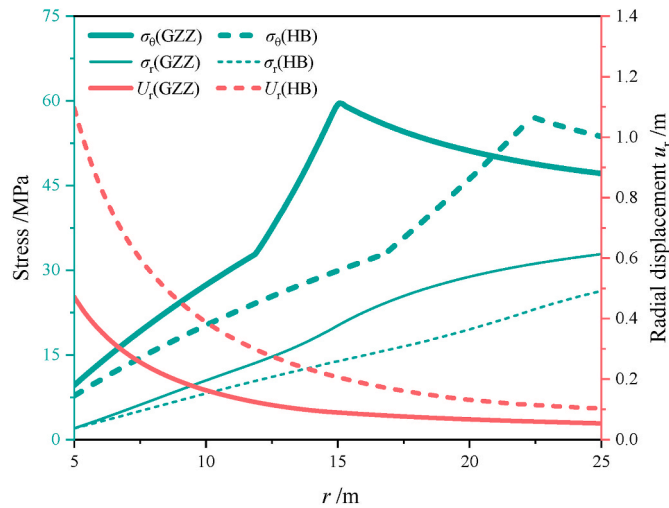
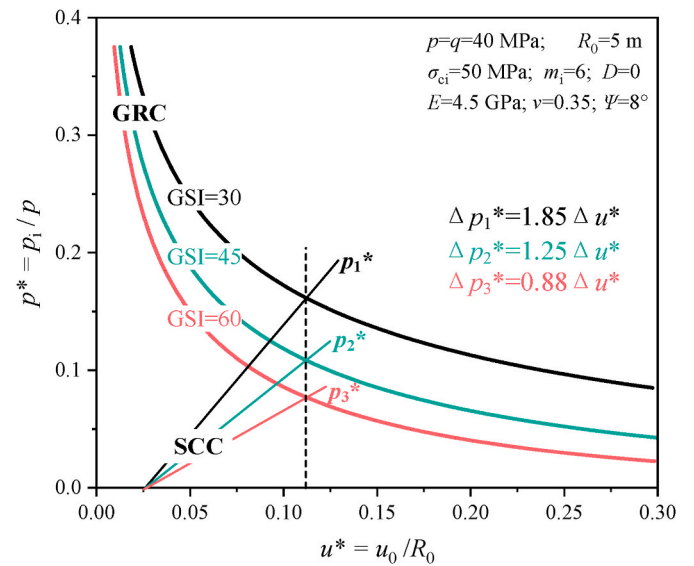


Fig. 13. Mechanical responses of surrounding rock using HB and GZZ strength criteria ($p = q = 40$ MPa).

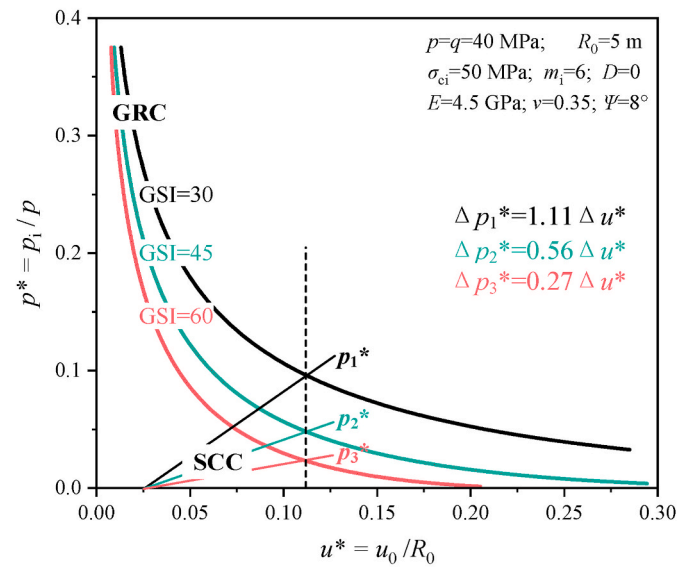
(3) There is no tedious and repetitive derivation process for a case-by-case analysis according to the relative values of principal stresses (σ_z , σ_r , and σ_θ) if smooth GZZ is used in 3D semi-analytical solution. The effect of σ_z (usually denoted as σ_2) on strength cannot be considered when HB is used for the semi-analytical solution, and the excavation response is significantly different from that of GZZ. The HB-based results are conservative, and may lead to waste of support materials and cause transitional support. Using an inappropriate strength criterion (e.g., 2D HB) in engineering design has a significant effect on proper selection of support measures, and its effect can even exceed the effect of misjudging the rock mass quality to a certain extent.

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(a) HB



(b) GZZ

Fig. 14. GRC at different GSIs using HB and GZZ.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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